44

Nuclear Physics

CHAPTER OUTLINE

44.1 Some Properties of Nuclei

44.2 Nuclear Binding Energy

44.3 Nuclear Models

44.4 Radioactivity

44.5 The Decay Processes

44.6 Natural Radioactivity

44.7 Nuclear Reactions

44.8 Nuclear Magnetic Resonance and Magnetic Resonance Imaging

* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ44.1 Answer (b). The frequency increases linearly with the magnetic field strength because the magnetic potential energy $-\vec{\boldsymbol{\mu}} \cdot \vec{\mathbf{B}}$ is proportional to the magnetic field strength.

OQ44.2 Answer (a). In the beta decay of ${}^{95}_{36}\text{Kr}$, the emitted particles are an electron, ${}^{0}_{-1}\text{e}$, and an antineutrino, $\bar{\nu}_{\text{e}}$. The emitted particles contain a total charge of $-e$ and zero nucleons. Thus, to conserve both charge and nucleon number, the daughter nucleus must be ${}^{95}_{37}\text{Rb}$, which contains $Z = 37$ protons and $A - Z = 95 - 37 = 58$ neutrons. (Recall that the electron and an antineutrino are produced by the decay on a neutron into a proton.)

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

OQ44.3 Answer (c). The emitted particle is not a nucleon because there is no change in nucleon number, and conservation of charge requires $15 = 16 + Z \rightarrow Z = -1$, so the emitted particle is an electron. From Equation 44.19, we see that ${}^{32}_{15}\mathrm{P}$ decays by means of beta decay: ${}^{32}_{15}\mathrm{P} \rightarrow {}^{32}_{16}\mathrm{S} + {}^{0}_{-1}\mathrm{e} + \bar{\nu}_{\mathrm{e}}$.

OQ44.4 Answer (d). In a large sample, one half of the radioactive nuclei initially present remain in the sample after one half-life has elapsed. Hence, the fraction of the original number of radioactive nuclei remaining after n half-lives have elapsed is $(1/2)^n = 1/2^n$. In this case the number of half-lives that have elapsed is $\Delta t/T_{1/2} = 14\ \mathrm{d}/3.6\ \mathrm{d} \approx 4$. Therefore, the approximate fraction of the original sample that remains undecayed is $1/2^4 = 1/16$.

OQ44.5 **(i)** Answer (b). Since the samples are of the same radioactive isotope, their half-lives are the same.

(ii) Answer (b). When prepared, sample G has twice the activity (number of radioactive decays per second) of sample H. The activity of a sample experiences exponential decay also; therefore, after 5 half-lives, the activity of sample G is decreased by a factor of 2^5, and after 5 half-lives the activity of sample H is decreased by a factor of 2^5. So after 5 half-lives, the ratio of activities is still 2:1.

OQ44.6 Answer (b). A gamma ray photon carries no nucleon number and no charge, so there can be no change in these quantities.

OQ44.7 Answer (c). The nucleus ${}^{40}_{18}\mathrm{X}$ contains $A = 40$ total nucleons, of which $Z = 18$ are protons. The remaining $A - Z = 40 - 18 = 22$ are neutrons.

OQ44.8 Answer (b). Conservation of nucleon number requires $144 = 140 + A \rightarrow A = 4$, and conservation of charge requires $60 = 58 + Z \rightarrow Z = 2$. The particle is ${}^{4}_{2}\mathrm{X} = {}^{4}_{2}\mathrm{He}$.

OQ44.9 Answer (d). The Q value for the reaction ${}^{9}_{4}\mathrm{Be} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{12}_{6}\mathrm{C} + {}^{1}_{0}\mathrm{n}$ is (using masses from Table 44.2)

$$\begin{aligned} Q = (\Delta m)c^2 &= \left(m_{{}^{9}_{4}\mathrm{Be}} + m_{{}^{4}_{2}\mathrm{He}} - m_{{}^{12}_{6}\mathrm{C}} - m_{\mathrm{n}}\right)c^2 \\ &= \left[9.012\,182\ \mathrm{u} + 4.002\,603\ \mathrm{u} \right. \\ &\qquad \left. -12.000\,000\ \mathrm{u} - 1.008\,665\ \mathrm{u}\right] \times \left(931.5\ \mathrm{Mev/u}\right) \\ &= 5.70\ \mathrm{MeV} \end{aligned}$$

OQ44.10 **(i)** Answer (a). The liquid drop model gives a simpler account of a nuclear fission reaction, including the energy released and the probable fission product nuclei.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(ii) Answer (b). The shell model predicts magnetic moments by necessarily describing the spin and orbital angular momentum states of the nucleons.

(iii) Answer (b). Again, the shell model wins when it comes to predicting the spectrum of an excited nucleus, as it allows only quantized energy states, and thus only specific transitions.

OQ44.11 Answer (d). A free neutron can undergo beta decay into a proton plus an electron and an antineutrino because its mass is greater than the mass of a free proton. Energy conservation prevents a free proton from decaying into a neutron plus a positron and a neutrino. (A proton bound inside a nucleus can undergo beta decay into a neutron if the final mass of the nucleus is less than that of the original nucleus, as for example in the beta decay of sodium-22: ${}^{22}_{11}\text{Na} \rightarrow \text{e}^{+} + \nu + {}^{22}_{10}\text{Ne}$.)

OQ44.12 Answer (d). The reaction energy is the amount of energy released as a result of a nuclear reaction. Equation 44.29 in the text implies that the reaction energy is (initial mass – final mass) c^2. The Q-value is taken as positive for an exothermic reaction.

OQ44.13 Answer (c). To conserve nucleon number (mass number), it is necessary that $A + 4 = 234$, or $A = 230$. Conservation of charge (atomic number) demands that $Z + 2 = 90$, or $Z = 88$.

ANSWERS TO CONCEPTUAL QUESTIONS

CQ44.1 The alpha particle and the daughter nucleus carry equal amounts of momentum in opposite directions. Since kinetic energy can be written as $\frac{p^2}{2m}$, the small-mass alpha particle has much more of the decay energy than the recoiling nucleus.

CQ44.2 The statement is false. Both patterns show monotonic decrease over time, but with very different shapes. For radioactive decay, maximum activity occurs at time zero. Cohorts of people now living will be dying most rapidly perhaps forty years from now. Everyone now living will be dead within less than two centuries, while the mathematical model of radioactive decay tails off exponentially forever. A radioactive nucleus never gets old. It has constant probability of decay however long it has existed.

CQ44.3 An alpha particle contains two protons and two neutrons. Because the nuclei of heavy hydrogen (D and T) contain only one proton, they cannot emit an alpha particle.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

CQ44.4 In alpha decay, there are only two final particles, the alpha particle and the daughter nucleus. There are also two conservation principles, energy and momentum, that apply to the process. As a result, the alpha particle must be ejected with a discrete energy to satisfy both conservation principles. Beta decay, however, is a three-particle decay involving the beta particle, the neutrino (or antineutrino), and the daughter nucleus. As a result, the energy and momentum can be shared in a variety of ways among the three particles while still satisfying the two conservation principles. This explains why the beta particle can have a continuous range of energies.

CQ44.5 Carbon dating cannot generally be used to estimate the age of a rock, because the rock was not alive to receive carbon, and hence radioactive carbon-14, from the environment. Only the ages of objects that were once alive can be estimated with carbon dating.

CQ44.6 The larger rest energy of the neutron means that a free proton in space will not spontaneously decay into a neutron and a positron. When the proton is in the nucleus, however, you must consider the total rest energy of the nucleus. If it is energetically favorable for the nucleus to have one fewer proton and one more neutron, then the process of positron decay will occur to achieve this lower energy.

CQ44.7 I refers to nuclear spin quantum number.

(a) I_z may have $2I+1=2\left(\frac{5}{2}\right)+1=6$ values for $I=\frac{5}{2}$, namely $\frac{5}{2}$, $\frac{3}{2}$, $\frac{1}{2}$, $-\frac{1}{2}$, $-\frac{3}{2}$, and $-\frac{5}{2}$.

(b) For $I = 3$, there are $2I + 1 = 2(3) + 1 = 7$ possible values for I_z.

CQ44.8 Extra neutrons are required to overcome the increasing electrostatic repulsion of the protons. The neutrons participate in the net attractive effect of the nuclear force, but feel no Coulomb repulsion.

CQ44.9 Nuclei with more nucleons than bismuth-209 are unstable because the electrical repulsion forces among all of the protons is stronger than the nuclear attractive force between nucleons.

CQ44.10 The nuclear force favors the formation of neutron-proton pairs, so a stable nucleus cannot be too far away from having equal numbers of protons and neutrons. This effect sets the upper boundary of the zone of stability on the neutron-proton diagram. All of the protons repel one another electrically, so a stable nucleus cannot have too many protons. This effect sets the lower boundary of the zone of stability.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

CQ44.11 Nucleus Y will be more unstable. The nucleus with the higher binding energy requires more energy to be disassembled into its constituent parts and has less available energy to release in a decay.

CQ44.12 After one half-life, one half the radioactive atoms have decayed. After the second half-life, one half of the remaining atoms have decayed. Therefore, $\frac{1}{2}+\frac{1}{4}=\frac{3}{4}$ of the original radioactive atoms have decayed after two half-lives.

CQ44.13 Long-lived progenitors at the top of each of the three natural radioactive series are the sources of our radium. As an example, thorium-232 with a half-life of 14 Gyr produces radium-228 and radium-224 at stages in its series of decays.

CQ44.14 Yes. The daughter nucleus can be left in its ground state or sometimes in one of a set of excited states. If the energy carried by the alpha particle is mysteriously low, the daughter nucleus can quickly emit the missing energy in a gamma ray.

CQ44.15 The alpha particle does not make contact with the nucleus because of electrostatic repulsion between the positively-charged nucleus and the $+2e$ alpha particle. To drive the alpha particle into the nucleus would require extremely high kinetic energy.

CQ44.16 The samples would have started with more carbon-14 than we first thought. We would increase our estimates of their ages.

CQ44.17 The photon and the neutrino are similar in that both particles have zero charge and little or no mass. (The photon has zero mass, but evidence suggests that neutrinos have a very small mass.) Both particles are capable of transferring both energy and momentum. They differ in that the photon has spin 1 and is involved in electromagnetic interactions, while the neutrino has spin $\frac{1}{2}$, interacts through the weak interaction, and is closely related to beta decay.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 44.1 Some Properties of Nuclei

P44.1 The average nuclear radii are $r = r_0 A^{1/3}$, where $r_0 = 1.2 \times 10^{-15}$ m = 1.2 fm and A is the mass number.

(a) For ${}_{1}^{2}\mathrm{H}$, $r = (1.2\text{ fm})(2)^{1/3} = \boxed{1.5\text{ fm}}$

(b) For ${}_{27}^{60}\mathrm{Co}$, $r = (1.2\text{ fm})(60)^{1/3} = \boxed{4.7\text{ fm}}$

(c) For ${}_{79}^{197}\mathrm{Au}$, $r = (1.2\text{ fm})(197)^{1/3} = \boxed{7.0\text{ fm}}$

(d) For ${}_{94}^{239}\mathrm{Pu}$, $r = (1.2\text{ fm})(239)^{1/3} = \boxed{7.4\text{ fm}}$

P44.2 (a) Approximate nuclear radii are given by $r = r_0 A^{1/3}$. Thus, if a nucleus of atomic number A has a radius approximately two-thirds that of ${}_{88}^{230}\mathrm{Ra}$, we should have

$$r = r_0 A^{1/3} = \frac{2}{3} r_0 (230)^{1/3}$$

or $$A = \frac{2^3}{3^3}(230) = \frac{8}{27}(230) \approx \boxed{68}$$

(b) $\boxed{\text{One possible nucleus is } {}_{30}^{68}\mathrm{Zn}}$.

(c) $\boxed{\begin{array}{l}\text{Isotopes of other elements to the left and right of zinc in}\\ \text{the periodic table (from manganese to bromine) may}\\ \text{have the same mass number.}\end{array}}$

P44.3 (a) The initial kinetic energy of the alpha particle must equal the electrostatic potential energy at the distance of closest approach.

$$K_i = U_f = \frac{k_e q Q}{r_{\min}}$$

$$r_{\min} = \frac{k_e q Q}{K_i} = \frac{\left(8.99 \times 10^9\ \mathrm{N \cdot m^2/C^2}\right)(2)(79)\left(1.60 \times 10^{-19}\ \mathrm{C}\right)^2}{(0.500\ \mathrm{MeV})\left(1.60 \times 10^{-13}\ \mathrm{J/MeV}\right)}$$

$$= 4.55 \times 10^{-13}\ \mathrm{m} = 455 \times 10^{-15}\ \mathrm{m} = \boxed{455\text{ fm}}$$

(b) Following the same logic as in part (a),

$$K_i = \frac{1}{2} m_\alpha v_i^2 = \frac{k_e q Q}{r_{\min}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Now, for $r_{\min} = 300 \text{ fm} = 300 \times 10^{-15} \text{ m}$, solving for the initial velocity gives

$$v_i = \sqrt{\frac{2k_e qQ}{m_\alpha r_{\min}}}$$

$$= \sqrt{\frac{2\left(8.99\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2\right)(2)(79)\left(1.602\times10^{-19}\ \text{C}\right)^2}{\left(6.645\times10^{-27}\ \text{kg}\right)\left(300\times10^{-15}\ \text{m}\right)}}$$

$$v_i = \boxed{6.05\times10^{6}\ \text{m/s}}$$

P44.4 An iron nucleus (in hemoglobin) has a few more neutrons than protons, but in a typical water molecule there are eight neutrons and ten protons. So protons and neutrons are nearly equally numerous in your body, each contributing mass (say) 35 kg:

(a) $35 \text{ kg}\left(\frac{1 \text{ nucleon}}{1.67\times10^{-27}\ \text{kg}}\right)\boxed{\sim 10^{28} \text{ protons}}$,

(b) and $\boxed{\sim 10^{28} \text{ neutrons}}$.

(c) The electron number is precisely equal to the proton number, $\boxed{\sim 10^{28} \text{ electrons}}$.

P44.5 (a) ${}^{65}_{29}\text{Cu}$ has an A number of 65, so the radius of its nucleus is

$$r = r_0 A^{1/3} = (1.2 \text{ fm})(65)^{1/3} = \boxed{4.8 \text{ fm}}$$

(b) The volume of the nucleus, assumed to be spherical in shape, is

$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi\left[r_0^3 A\right] = \frac{4}{3}\pi\left[\left(1.2\times10^{-15}\ \text{m}\right)^3(65)\right]$$

$$= \boxed{4.7\times10^{-43}\ \text{m}^3}$$

(c) The density of the nucleus is

$$\rho = \frac{m}{V} = \frac{Am}{\frac{4}{3}\pi\left[r_0^3 A\right]} = \frac{3m}{4\pi r_0^3} = \frac{3\left(1.66\times10^{-27}\ \text{kg}\right)}{4\pi\left(1.2\times10^{-15}\ \text{m}\right)^3}$$

$$= \boxed{2.3\times10^{17}\ \text{kg/m}^3}$$

P44.6 From $M_E = \rho_n V = \rho_n (4\pi r^3/3)$, we find

$$r = \left(\frac{3M_E}{4\pi\rho_n}\right)^{1/3} = \left[\frac{3\left(5.98\times10^{24}\ \text{kg}\right)}{4\pi\left(2.30\times10^{17}\ \text{kg/m}^3\right)}\right]^{1/3} = \boxed{184 \text{ m}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P44.7 The number of neutrons in a star of two solar masses is

$$A = \frac{2\left(1.99\times10^{30}\ \text{kg}\right)}{1.67\times10^{-27}\ \text{kg/neutron}} = 2.38\times10^{57}\ \text{neutrons}$$

Therefore,

$$r = r_0 A^{1/3} = \left(1.20\times10^{-15}\ \text{m}\right)\left(2.38\times10^{57}\right)^{1/3}$$
$$= 1.6\times10^{4}\ \text{m} = \boxed{16\ \text{km}}$$

P44.8 (a) The electric potential energy between two protons is

$$U = k_e \frac{q_1 q_2}{r} = k_e \frac{e^2}{r}$$
$$= \left(8.99\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\left[\frac{\left(1.60\times10^{-19}\ \text{C}\right)^2}{4.00\times10^{-15}\ \text{m}}\right]$$
$$\times\left(\frac{1\ \text{eV}}{1.60\times10^{-19}\ \text{J}}\right)\left(\frac{1\ \text{MeV}}{10^{6}\ \text{eV}}\right)$$
$$= \boxed{0.360\ \text{MeV}}$$

(b) Figure P44.8 shows the highest point in the curve at about 4 MeV, a factor of ten higher than the value in (a).

P44.9 By energy conservation,

$$\frac{1}{2}mv^2 = q\Delta V: \qquad 2m\Delta V = qr^2B^2$$

By Newton's second law,

$$\frac{mv^2}{r} = qvB: \qquad r = \sqrt{\frac{2m\Delta V}{qB^2}}$$

Comparing radii for particles with different masses but with the same charge, we find that

$$\frac{r_2}{r_1} = \frac{\sqrt{2m_2\Delta V/qB^2}}{\sqrt{2m_1\Delta V/qB^2}} = \frac{\sqrt{m_2}}{\sqrt{m_1}}$$

For ^{12}C: $m_1 = 12$ u and $r_1 = 7.89$ cm

For ^{13}C:

$$\frac{r_2}{r_1} = \frac{r_2}{7.89\ \text{cm}} = \frac{\sqrt{m_2}}{\sqrt{m_1}} = \frac{\sqrt{13}}{\sqrt{12}} \qquad \rightarrow \qquad \boxed{8.21\ \text{cm}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P44.10 By energy conservation,

$$\frac{1}{2}mv^2 = q\Delta V: \qquad 2m\Delta V = qr^2B^2$$

By Newton's second law,

$$\frac{mv^2}{r} = qvB: \qquad r = \sqrt{\frac{2m\Delta V}{qB^2}}$$

Comparing radii for particles with different masses but with the same charge, we find that

$$\frac{r_2}{r_1} = \frac{\sqrt{2m_2\Delta V/qB^2}}{\sqrt{2m_1\Delta V/qB^2}} = \frac{\sqrt{m_2}}{\sqrt{m_1}} \qquad \rightarrow \qquad \boxed{r_2 = \sqrt{\frac{m_2}{m_1}}r_1}$$

***P44.11** (a) The magnitude of the maximum Coulomb force is given by

$$F_{\max} = \frac{k_e q_1 q_2}{r_{\min}^2}$$

$$= \frac{\left(8.99\times10^9\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\left[(2)(6)\left(1.60\times10^{-19}\ \text{C}\right)^2\right]}{\left(1.00\times10^{-14}\ \text{m}\right)^2}$$

$$= \boxed{27.6\ \text{N}}$$

(b) From Newton's second law,

$$a_{\max} = \frac{F_{\max}}{m_\alpha} = \frac{27.6\ \text{N}}{6.64\times10^{-27}\ \text{kg}} = \boxed{4.16\times10^{27}\ \text{m/s}^2}$$

(c) The potential energy of the system at the time of the maximum force is

$$U_{\max} = \frac{k_e q_1 q_2}{r_{\min}}$$

$$= \left\{\frac{\left(8.99\times10^9\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\left[(2)(6)\left(1.60\times10^{-19}\ \text{C}\right)^2\right]}{\left(1.00\times10^{-14}\ \text{m}\right)}\right\}$$

$$\times\left(\frac{1\ \text{eV}}{1.60\times10^{-19}\ \text{J}}\right)\left(\frac{1\ \text{MeV}}{10^6\ \text{eV}}\right)$$

$$= \boxed{1.73\ \text{MeV}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P44.12 We obtain the alpha particle's momentum from

$$E_\alpha = 7.70 \text{ MeV} = \frac{1}{2}mv^2 = \frac{1}{2}\frac{(mv)^2}{m} \qquad \rightarrow \qquad mv = \sqrt{2mE_\alpha}$$

(a) The de Broglie wavelength of the alpha particle is (mass from Table 44.1)

$$\lambda = \frac{h}{m_\alpha v_\alpha} = \frac{h}{\sqrt{2m_\alpha E_\alpha}}$$

$$= \frac{6.626\times10^{-34}\ \text{J}\cdot\text{s}}{\sqrt{2\left(6.64\times10^{-27}\ \text{kg}\right)\left(7.70\times10^{6}\ \text{eV}\right)\left(1.60\times10^{-19}\ \text{J/eV}\right)}}$$

$$= 5.18\times10^{-15}\ \text{m} = \boxed{5.18\ \text{fm}}$$

(b) Since $\boxed{\lambda \text{ is much less than the distance of closest approach,}}$ the alpha particle may be considered a particle.

P44.13 The volume of each of the golf balls is

$$V = \frac{4}{3}\pi r^4 = \frac{4}{3}\pi\left(0.021\,5\ \text{m}\right)^3 = 4.16\times10^{-5}\ \text{m}^3$$

We take the nuclear density from Example 44.2. Then, the mass of a golf-ball sized nuclear matter is

$$m = \rho V = \left(2.3\times10^{17}\ \text{kg/m}^3\right)\left(4.16\times10^{-5}\ \text{m}^3\right) = 9.6\times10^{12}\ \text{kg}$$

and the gravitational force between two such balls is

$$F = G\frac{m_1 m_2}{r^2} = \left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\frac{\left(9.6\times10^{12}\ \text{kg}\right)^2}{\left(1.00\ \text{m}\right)^2}$$

$$F = \boxed{6.1\times10^{15}\ \text{N toward each other.}}$$

P44.14 (a) Let V represent the volume of the tank. The number of molecules present is

$$N = nN_A = \frac{PV}{RT} = \frac{\left(1.013\times10^{5}\ \text{N/m}^2\right)V}{(8.315\ \text{J/mol}\cdot\text{K})(273\ \text{K})}\left(6.022\times10^{23}\right)$$

$$= \left(2.69\times10^{25}\ \text{m}^{-3}\right)V$$

The volume of one molecule is

$$2\left(\frac{4}{3}\pi r^3\right) = \frac{8\pi}{3}\left(\frac{1.00\times10^{-10}\ \text{m}}{2}\right)^3 = 1.047\times10^{-30}\ \text{m}^3$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

The volume of all the molecules is

$$\left(2.69\times10^{25}\ \text{m}^{-3}\right)V\left(1.047\times10^{-30}\ \text{m}^{3}\right)=2.82\times10^{-5}\ V$$

So the fraction of the volume occupied by the hydrogen molecules is $\boxed{2.82\times10^{-5}.}$ An atom is precisely one half of a molecule.

(b) The fraction occupied by the nucleus is found from

$$\frac{\text{nuclear volume}}{\text{atomic volume}}=\frac{\frac{4}{3}\pi r^{3}}{\frac{4}{3}\pi(d/2)^{3}}=\left(\frac{r}{d/2}\right)^{3}$$

$$=\left(\frac{1.20\times10^{-15}\ \text{m}}{0.500\times10^{-10}\ \text{m}}\right)^{3}=\boxed{1.38\times10^{-14}}$$

In linear dimension, the nucleus is small inside the atom in the way a fat strawberry is small inside the width of the Grand Canyon. In terms of volume, the nucleus is *really* small.

Section 44.2 Nuclear Binding Energy

P44.15 Using Equation 44.2, the binding energy per nucleon is

$$\frac{E_b}{A}=\frac{\left[ZM(\text{H})+Nm_n-M\left({}^{A}_{Z}\text{X}\right)\right]}{A}\left(\frac{931.5\ \text{MeV}}{\text{u}}\right)$$

Using atomic masses as given in Table 44.2,

(a) For ${}^{2}_{1}\text{H}$:

$$\frac{E_b}{A}=\frac{1(1.007\ 825\ \text{u})+1(1.008\ 665\ \text{u})-2.014\ 102\ \text{u}}{2}$$

$$=\left(\frac{0.002\ 388\ \text{u}}{2}\right)\left(\frac{931.5\ \text{MeV}}{\text{u}}\right)=\boxed{1.11\ \text{MeV}}$$

(b) For ${}^{4}_{2}\text{He}$:

$$\frac{E_b}{A}=\frac{2(1.007\ 825\ \text{u})+2(1.008\ 665\ \text{u})-4.002\ 603\ \text{u}}{4}$$

$$=\left(\frac{0.030\ 377\ \text{u}}{4}\right)\left(\frac{931.5\ \text{MeV}}{\text{u}}\right)=\boxed{7.07\ \text{MeV}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(c) For $^{56}_{26}\text{Fe}$:

$$\frac{E_b}{A} = \frac{26(1.007\ 825\ \text{u}) + 30(1.008\ 665\ \text{u}) - 55.934\ 942\ \text{u}}{56}$$

$$= \left(\frac{0.528\ 458\ \text{u}}{56}\right)\left(\frac{931.5\ \text{MeV}}{\text{u}}\right) = \boxed{8.79\ \text{MeV}}$$

(d) For $^{238}_{92}\text{U}$:

$$\frac{E_b}{A} = \frac{92(1.007\ 825\ \text{u}) + 146(1.008\ 665\ \text{u}) - 238.050\ 783\ \text{u}}{238}$$

$$= \left(\frac{1.934\ 207\ \text{u}}{238}\right)\left(\frac{931.5\ \text{MeV}}{\text{u}}\right) = \boxed{7.57\ \text{MeV}}$$

P44.16 We use Equation 44.2,

$$E_b(\text{MeV}) = \left[ZM(\text{H}) + Nm_n - M\left({}^{A}_{Z}\text{X}\right)\right](931.494\ \text{MeV/u})$$

Then, for $^{23}_{11}\text{Na}$,

$$E_b\left({}^{23}_{11}\text{Na}\right) = \left[11M(\text{H}) + 12m_n - M\left({}^{23}_{11}\text{Na}\right)\right](931.494\ \text{MeV/u})$$

$$= \left[11(1.007\ 825\ \text{u}) + 12(1.008\ 665\ \text{u}) - 22.989\ 769\ \text{u}\right] \times (931.494\ \text{MeV/u})$$

$$= 186.565\ \text{MeV}$$

and $$\frac{E_b}{A} = \frac{186.565\ \text{MeV}}{23} = 8.11\ \text{MeV}$$

For $^{23}_{12}\text{Mg}$,

$$E_b = E_b\left({}^{23}_{12}\text{Mg}\right)$$

$$= \left[12M(\text{H}) + 11m_n - M\left({}^{23}_{12}\text{Mg}\right)\right](931.494\ \text{MeV/u})$$

$$= \left[12(1.007\ 825\ \text{u}) + 11(1.008\ 665\ \text{u}) - 22.994\ 124\ \text{u}\right] \times (931.494\ \text{MeV/u})$$

$$= 181.726\ \text{MeV}$$

and $$\frac{E_b}{A} = \frac{181.726\ \text{MeV}}{23} = 7.90\ \text{MeV}$$

The difference is

$$\frac{\Delta E_b}{A} = \frac{E_b\left({}^{23}_{11}\text{Na}\right) - E_b\left({}^{23}_{12}\text{Mg}\right)}{A}$$

$$= 8.11\ \text{MeV} - 7.90\ \text{MeV} = 0.210\ \text{MeV}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

The binding energy per nucleon is greater for $^{23}_{11}\text{Na}$ by $\boxed{0.210\ \text{MeV}}$.
There is less proton repulsion in $^{23}_{11}\text{Na}$; it is the more stable nucleus.

P44.17 From Equation 44.2, the binding energy of a nucleus is

$$E_b(\text{MeV}) = \left[ZM(\text{H}) + Nm_n - M\left({}^{A}_{Z}\text{X}\right)\right](931.494\ \text{MeV/u})$$

For $^{15}_{8}\text{O}$:

$$E_b = \left[8(1.007\ 825\ \text{u}) + 7(1.008\ 665\ \text{u}) - 15.003\ 065\ \text{u}\right] \times (931.494\ \text{MeV/u}) = 111.96\ \text{MeV}$$

For $^{15}_{7}\text{N}$:

$$E_b = \left[7(1.007\ 825\ \text{u}) + 8(1.008\ 665\ \text{u}) - 15.000\ 109\ \text{u}\right] \times (931.494\ \text{MeV/u}) = 115.49\ \text{MeV}$$

Therefore, the binding energy is $\boxed{\text{greater for } {}^{15}_{7}\text{N} \text{ by } 3.54\ \text{MeV}.}$

P44.18 We find the mass difference, $\Delta M = Zm_{\text{H}} + Nm_{\text{n}} - M$, and then the binding energy per nucleon, $\dfrac{E_b}{A} = \dfrac{\Delta M(931.5)}{A}$, in units of MeV. The results are tabulated below.

Nuclei	Z	N	M in u	ΔM in u	$\dfrac{E_b}{A}$ in MeV
^{55}Mn	25	30	54.938 050	0.517 5	8.765
^{56}Fe	26	30	55.934 942	0.528 46	8.790
^{59}Co	27	32	58.933 200	0.555 35	8.768

$\boxed{\therefore\ {}^{56}\text{Fe} \text{ has a greater } \dfrac{E_b}{A} \text{ than its neighbors.}}$

P44.19 (a) The isobar with the highest neutron-to-proton ratio is $\boxed{{}^{139}_{55}\text{Cs}}$; the ratio is $\dfrac{N}{Z} = \dfrac{A-Z}{Z} = \dfrac{139-55}{55} = \dfrac{84}{55} = 1.53$

(b) $\boxed{{}^{139}_{57}\text{La}}$ is stable, so has the largest binding energy per nucleon (8.378 MeV).

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(c) The isobars are close in Figure 44.6, the plot of binding energy per nucleon versus mass number, and there is not much detail, so we may assume they have about the same binding energy, or missing mass. However, neutrons have more mass than protons, so the isobar with more neutrons (thus, fewer protons) should be more massive: $\boxed{^{139}_{55}\text{Cs}}$.

P44.20 (a) The radius of the ^{40}Ca nucleus is,

$$R = r_0 A^{1/3} = \left(1.20 \times 10^{-15}\ \text{m}\right)(40)^{1/3} = 4.10 \times 10^{-15}\ \text{m}$$

The energy required to overcome electrostatic repulsion is

$$U = \frac{3k_e Q^2}{5R} = \frac{3\left(8.99 \times 10^9\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\left[20\left(1.602 \times 10^{-19}\ \text{C}\right)\right]^2}{5\left(4.10 \times 10^{-15}\ \text{m}\right)}$$

$$= 1.35 \times 10^{-11}\ \text{J} = \boxed{84.2\ \text{MeV}}$$

(b) The binding energy of $^{40}_{20}\text{Ca}$ ($Z = 20$, $N = A - Z = 20$) is (using Equation 44.2 and masses from Table 44.2),

$$E_b = \left[20\left(1.007\,825\ \text{u}\right) + 20\left(1.008\,665\ \text{u}\right) - 39.962\,591\ \text{u}\right] \times \left(931.5\ \text{MeV/u}\right)$$

$$= \boxed{342\ \text{MeV}}$$

(c) The nuclear force is so strong that the binding energy greatly exceeds the minimum energy needed to overcome electrostatic repulsion.

P44.21 Removal of a neutron from $^{43}_{20}\text{Ca}$ would result in the residual nucleus, $^{42}_{20}\text{Ca}$. If the required separation energy is ΔE_n, the overall process can be described by

$$\text{mass}\left(^{43}_{20}\text{Ca}\right) + \Delta E_\text{n} = \text{mass}\left(^{42}_{20}\text{Ca}\right) + \text{mass}(\text{n})$$

$$\Delta E_\text{n} = \text{mass}\left(^{42}_{20}\text{Ca}\right) + \text{mass}(\text{n}) - \text{mass}\left(^{43}_{20}\text{Ca}\right)$$

From Table 44.2,

$$\Delta E_\text{n} = \left(41.958\,618\ \text{u} + 1.008\,665\ \text{u} - 42.958\,767\ \text{u}\right) \times \left(931.5\ \text{MeV/u}\right)$$

$$= \boxed{7.93\ \text{MeV}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Section 44.3 Nuclear Models

P44.22 The curve of binding energy shows that a heavy nucleus of mass number $A = 200$ has binding energy about

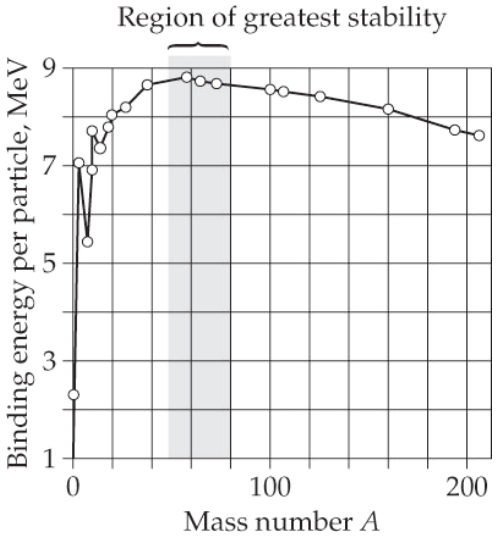
$$\left(7.8\frac{\text{MeV}}{\text{nucleon}}\right)(200 \text{ nucleons}) \approx 1.56 \text{ GeV}$$

Thus, it is less stable than its potential fission products, two middleweight nuclei of $A = 100$, together having binding energy

$$2(8.7 \text{ MeV/nucleon})(100 \text{ nucleons}) \approx 1.74 \text{ GeV}$$

Fission then releases about

$$1.74 \text{ GeV} - 1.56 \text{ GeV} \quad \boxed{\sim 200 \text{ MeV}}$$



ANS. FIG. P44.22

P44.23 (a) In Equation 44.3, the first or "Volume" term is,

$$E_1 = C_1 A = (15.7 \text{ MeV})(56) = 879 \text{ MeV}$$

The second, or "Surface" term is,

$$E_2 = -C_2 A^{2/3} = -(17.8 \text{ MeV})(56)^{2/3} = -260 \text{ MeV}$$

The third, or "Coulomb" term is,

$$E_3 = -C_3 \frac{Z(Z-1)}{A^{1/3}} = -(0.71 \text{ MeV})\frac{(26)(25)}{(56)^{1/3}}$$

$$= -121 \text{ MeV}$$

and the fourth, or "Asymmetry" term is,

$$E_4 = C_4 \frac{(A-2Z)^2}{A} = -(23.6 \text{ MeV})\frac{(56-52)^2}{56} = -6.74 \text{ MeV}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

The binding energy is then

$$E_b = C_1 A - C_2 A^{2/3} - C_3 \frac{Z(Z-1)}{A^{1/3}} - C_4 \frac{(A-2Z)^2}{A}$$

$$= 879 \text{ MeV} - 260 \text{ MeV} - 121 \text{ MeV} - 6.74 \text{ MeV} = \boxed{491 \text{ MeV}}$$

(b) The percentages for each of the terms is as follows

$$\boxed{\text{term 1}:} \frac{E_1}{E_b} = \boxed{179\%}; \quad \boxed{\text{term 2}:} \frac{E_2}{E_b} = \boxed{-53.0\%};$$

$$\boxed{\text{term 3:}} \frac{E_3}{E_b} = \boxed{-24.6\%}; \quad \boxed{\text{term 4:}} \frac{E_4}{E_b} = \boxed{-1.37\%}$$

P44.24 (a) Nucleons on the surface have fewer neighbors with which to interact. The surface term is negative to reduce the estimate from the volume term, which assumes that all nucleons have the same number of neighbors.

(b) The volume to surface ratio for a sphere of radius r is

$$\frac{\text{Volume}}{\text{Area}} = \frac{(4/3)\pi r^3}{4\pi r^2} = \boxed{\frac{1}{3} r}$$

The volume to surface ratio for a cube of side length L is

$$\frac{\text{Volume}}{\text{Area}} = \frac{L^3}{6L^2} = \boxed{\frac{1}{6} L}$$

The sphere has a larger ratio to its characteristic length, so it would represent a larger binding energy and be more plausible for a nuclear shape.

Section 44.4 Radioactivity

***P44.25** We use Equation 44.7 for the exponential decay rate of the sample, $R = R_0 e^{-\lambda t}$, where

$$\lambda = \frac{\ln 2}{26.0 \text{ h}} = 0.026\ 7 \text{ h}^{-1}$$

Since we require a 90% decrease in activity,

$$\frac{R}{R_0} = 0.100 = e^{-\lambda t} \quad \rightarrow \quad \ln(0.100) = -\lambda t$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

then,

$$t = \frac{2.30}{0.026\ 7/\text{h}} = \boxed{86.4\ \text{h}}$$

P44.26 (a) From $R = R_0 e^{-\lambda t}$, the decay constant is

$$\lambda = \frac{1}{t}\ln\left(\frac{R_0}{R}\right) = \left(\frac{1}{4.00\ \text{h}}\right)\ln\left(\frac{10.0\ \text{mCi}}{8.00\ \text{mCi}}\right) = 5.58\times10^{-2}\ \text{h}^{-1}$$

$$= \boxed{1.55\times10^{-5}\ \text{s}^{-1}}$$

(b) The half-life is

$$T_{1/2} = \frac{\ln 2}{\lambda} = \boxed{12.4\ \text{h}}$$

(c) The number of original atoms can be found if we convert the initial activity from curies into becquerels (decays per second): 1 Ci $\equiv$ 3.70 $\times$ l0^{10} Bq.

$$R_0 = 10.0\ \text{mCi} = \left(10.0\times10^{-3}\ \text{Ci}\right)\left(3.70\times10^{10}\ \text{Bq/Ci}\right)$$

$$= 3.70\times10^{8}\ \text{Bq}$$

Since $R_0 = \lambda N_0$, the original number of nuclei is

$$N_0 = \frac{R_0}{\lambda} = \frac{3.70\times10^{8}\ \text{decays/s}}{1.55\times10^{-5}\ \text{s}} = \boxed{2.39\times10^{13}\ \text{atoms}}$$

(d) The decay rate after thirty hours is

$$R = R_0 e^{-\lambda t} = (10.0\ \text{mCi})\exp\left[\left(-5.58\times10^{-2}\ \text{h}^{-1}\right)(30.0\ \text{h})\right]$$

$$= \boxed{1.88\ \text{mCi}}$$

P44.27 The decay law is

$$dN/dt = -\lambda N$$

Then, the decay constant is

$$\lambda = -\frac{1}{N}\left(\frac{dN}{dt}\right) = -\left(\frac{1}{1.00\times10^{15}\ \text{nuclei}}\right)\left(\frac{-6.00\times10^{11}\ \text{nuclei}}{\text{s}}\right)$$

$$= 6.00\times10^{-4}\ \text{s}^{-1}$$

and the half-life is

$$T_{1/2} = \frac{\ln 2}{\lambda} = \boxed{1.16\times10^{3}\ \text{s}}$$

(This is 19.3 minutes.)

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P44.28 According to Equation 44.7, the time dependence of the decay rate is $R = R_0 e^{-\lambda \Delta t}$. From this equation we can derive a relation between the change in decay rate over the time interval Δt to the decay constant. We start with $R = R_0 e^{-\lambda \Delta t}$. Then, rearranging and taking the natural log of both sides gives

$$e^{-\lambda \Delta t} = \frac{R}{R_0} \qquad \rightarrow \qquad \ln\left(e^{-\lambda \Delta t}\right) = \ln\left(\frac{R}{R_0}\right)$$

or $$-\lambda \Delta t = \ln\left(\frac{R}{R_0}\right) = -\ln\left(\frac{R_0}{R}\right)$$

Solving,

$$\lambda = \frac{1}{\Delta t}\ln\left(\frac{R_0}{R}\right)$$

Now, because $\lambda = \dfrac{\ln 2}{T_{1/2}}$, we can relate the time interval Δt to the half-life:

$$\lambda = \frac{1}{\Delta t}\ln\left(\frac{R_0}{R}\right) \qquad \rightarrow \qquad \frac{\ln 2}{T_{1/2}} = \frac{1}{(\ln 2)\Delta t}\ln\left(\frac{R_0}{R}\right)$$

$$\frac{1}{T_{1/2}} = \frac{1}{(\ln 2)\Delta t}\ln\left(\frac{R_0}{R}\right)$$

$$T_{1/2} = \frac{(\ln 2)\Delta t}{\ln\left(R_0/R\right)}$$

P44.29 The number of nuclei that decay during the interval will be

$$\Delta N = N_1 - N_2 = N_0\left(e^{-\lambda t_1} - e^{-\lambda t_2}\right)$$

First we find the decay constant λ:

$$\lambda = \frac{\ln 2}{T_{1/2}} = \frac{0.693}{64.8\ \text{h}} = 0.010\ 7\ \text{h}^{-1} = 2.97 \times 10^{-6}\ \text{s}^{-1}$$

Now we find N_0:

$$N_0 = \frac{R_0}{\lambda} = \frac{(40.0\ \mu\text{Ci})}{2.97 \times 10^{-6}\ \text{s}^{-1}}\left(\frac{3.70 \times 10^{4}\ \text{s}^{-1}}{\mu\text{Ci}}\right)$$

$$= 4.98 \times 10^{11}\ \text{nuclei}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Substituting in these values,

$$N_1 - N_2 = \left(4.98 \times 10^{11}\right)\left[e^{-(\ln 2/64.8\ \text{h})(10.0\ \text{h})} - e^{-(\ln 2/64.8\ \text{h})(12.0\ \text{h})}\right]$$

$$N_1 - N_2 = \boxed{9.47 \times 10^{9}\ \text{nuclei}}$$

P44.30 The number of nuclei that decay during the interval will be

$$N_1 - N_2 = N_0\left(e^{-\lambda t_1} - e^{-\lambda t_2}\right)$$

We wish to write this expression in terms of the half-life $T_{1/2}$ and the initial decay rate R_0. First, from the definition of λ, we have

$$\lambda = \frac{\ln 2}{T_{1/2}} \quad \rightarrow \quad e^{-\lambda t} = e^{\ln 2\left(-t/T_{1/2}\right)} = 2^{-t/T_{1/2}}$$

Now we find N_0:

$$N_0 = \frac{R_0}{\lambda} = \frac{R_0 T_{1/2}}{\ln 2}$$

Substituting in these expressions, we find that

$$N_1 - N_2 = \frac{R_0 T_{1/2}}{\ln 2}\left(e^{-\lambda t_1} - e^{-\lambda t_2}\right) = \boxed{\frac{R_0 T_{1/2}}{\ln 2}\left(2^{-t_1/T_{1/2}} - 2^{-t_2/T_{1/2}}\right)}$$

P44.31 (a) The decay constant is

$$\lambda = \frac{\ln 2}{T_{1/2}} = \frac{\ln 2}{8.04\ \text{d}} = \boxed{0.0862\ \text{d}^{-1}}$$

$$= \frac{0.0862}{\text{d}}\left(\frac{1\ \text{d}}{24\ \text{h}}\right) = \boxed{3.59 \times 10^{-3}\ \text{h}^{-1}}$$

$$= \frac{9.98 \times 10^{-7}}{\text{h}}\left(\frac{1\ \text{h}}{3600\ \text{s}}\right) = \boxed{9.98 \times 10^{-7}\ \text{s}^{-1}}$$

(b) From $R = \lambda N$, the number of radioactive nuclei in a 6.40 mCi of ^{131}I is

$$N = \frac{R}{\lambda} = \frac{6.40 \times 10^{-3}\ \text{Ci}}{9.98 \times 10^{-7}\ \text{s}^{-1}}\left(\frac{3.70 \times 10^{10}\ \text{s}^{-1}}{\text{Ci}}\right) = \boxed{2.37 \times 10^{14}\ \text{nuclei}}$$

(c) From Equation 44.7, $R = \lambda N$, the decay rate R also undergoes exponential decay; thus, after one half-life, the rate drops from R_0 to $R_0/2$. The number of half-lives that have elapsed after 40.2 d is $n = t/T_{1/2} = 40.2\ \text{d}/8.04\ \text{d} = 5$, so the remaining activity of the sample is

$$R = \frac{R_0}{2^n} = \frac{R_0}{2^5} = \frac{6.40\ \text{mCi}}{32} = \boxed{0.200\ \text{mCi}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P44.32 (a) From Equation 44.6, the fraction remaining at $t = 5.00$ yr will be

$$\frac{N}{N_0} = e^{-\lambda t} = e^{-t \ln 2/T_{1/2}} = e^{-(5.00 \text{ yr}) \ln 2/(12.33 \text{ yr})} = \boxed{0.755}$$

(b) At $t = 10.0$ yr,

$$\frac{N}{N_0} = e^{-\lambda t} = e^{-t \ln 2/T_{1/2}} = e^{-(10.0 \text{ yr}) \ln 2/(12.33 \text{ yr})} = \boxed{0.570}$$

(c) At $t = 123.3$ yr,

$$\frac{N}{N_0} = e^{-\lambda t} = e^{-t \ln 2/T_{1/2}} = e^{-(123.3 \text{ yr}) \ln 2/(12.33 \text{ yr})} = e^{-10 \ln 2} = \boxed{9.766 \times 10^{-4}}$$

(d) No. The decay model depends on large numbers of nuclei. After some long but finite time, only one undecayed nucleus will remain. It is likely that the decay of this final nucleus will occur before infinite time.

P44.33 The number remaining after time $\frac{T_{1/2}}{2} = \frac{\ln 2}{2\lambda}$ is

$$N = N_0 e^{-\lambda t} = N_0 \, e^{-\lambda(\ln 2/2\lambda)} = N_0 \left(e^{-\ln 2}\right)^{1/2} = N_0 \left(\frac{1}{2}\right)^{1/2} = \frac{N_0}{\sqrt{2}}$$

The number decaying in this first half of the first half-life is

$$\Delta N_{\text{first half}} = N_0 - \frac{N_0}{\sqrt{2}} = \left(1 - \frac{1}{\sqrt{2}}\right) N_0 = \left(1 - \frac{\sqrt{2}}{2}\right) N_0$$

$$= \frac{\sqrt{2}}{2}\left(\sqrt{2} - 1\right) N_0$$

The number remaining after time $T_{1/2}$ is $\frac{N_0}{2}$, so the number decaying in the second half of the first half-life is

$$\Delta N_{\text{second half}} = \frac{N_0}{\sqrt{2}} - \frac{N_0}{2} = \left(\frac{1}{\sqrt{2}} - \frac{1}{2}\right) N_0 = \left(\frac{\sqrt{2}}{2} - \frac{1}{2}\right) N_0$$

$$= \frac{1}{2}\left(\sqrt{2} - 1\right) N_0$$

The ratio required is then

$$\frac{\Delta N_{\text{first half}}}{\Delta N_{\text{second half}}} = \frac{\frac{\sqrt{2}}{2}\left(\sqrt{2} - 1\right) N_0}{\frac{1}{2}\left(\sqrt{2} - 1\right) N_0} = \sqrt{2} = \boxed{1.41}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P44.34 (a) $\dfrac{dN_2}{dt} =$ rate of change of N_2

= rate of production of N_2 – rate of decay of N_2

= rate of decay of N_1 – rate of decay of N_2

$= \lambda_1 N_1 - \lambda_2 N_2$

(b) From the trial solution,

$$N_2(t) = \frac{N_{10}\lambda_1}{\lambda_1 - \lambda_2}\left(e^{-\lambda_2 t} - e^{-\lambda_1 t}\right)$$

$$\therefore \frac{dN_2}{dt} = \frac{N_{10}\lambda_1}{\lambda_1 - \lambda_2}\left(-\lambda_2 e^{-\lambda_2 t} + \lambda_1 e^{-\lambda_1 t}\right) \qquad \textbf{[1]}$$

$$\frac{dN_2}{dt} + \lambda_2 N_2 = \frac{N_{10}\lambda_1}{\lambda_1 - \lambda_2}\left(\cancel{-\lambda_2 e^{-\lambda_2 t}} + \lambda_1 e^{-\lambda_1 t}\right) + \frac{N_{10}\lambda_1}{\lambda_1 - \lambda_2}\left(\cancel{\lambda_2 e^{-\lambda_2 t}} - \lambda_2 e^{-\lambda_1 t}\right)$$

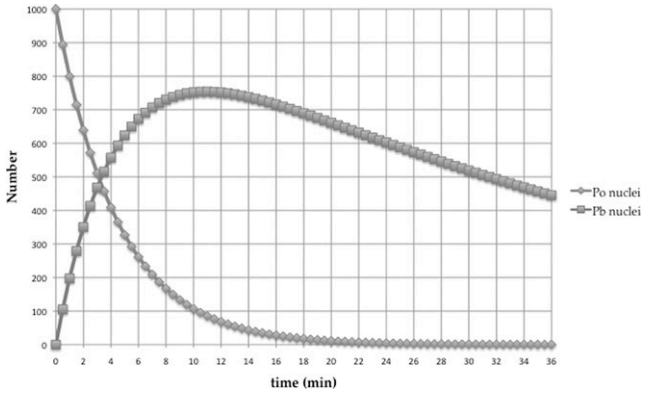
$$= \frac{N_{10}\lambda_1}{\lambda_1 - \lambda_2}\left(\lambda_1 - \lambda_2\right)e^{-\lambda_1 t} = \lambda_1 N_1$$

So $\dfrac{dN_2}{dt} = \lambda_1 N_1 - \lambda_2 N_2$ as required.

(c) The functions plotted in ANS. FIG. P44.34(c) are

Po nuclei: $N_1(t) = 1\,000 e^{-(\ln 2/3.10\ \text{min})t}$

Pb nuclei: $N_2(t) = 1\,130.8\left[e^{-(\ln 2/26.8\ \text{min})t} - e^{-(\ln 2/3.10\ \text{min})t}\right]$



ANS. FIG. P44.34(c)

(d) From the graph, $t_m \approx \boxed{10.9\ \text{min}}$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(e) From equation [1], $\frac{dN_2}{dt} = 0$ if

$$\lambda_2 e^{-\lambda_2 t} = \lambda_1 e^{-\lambda_1 t}$$

$$\therefore e^{(\lambda_1 - \lambda_2)t} = \frac{\lambda_1}{\lambda_2}$$

Thus, $t = \boxed{t_m = \frac{\ln(\lambda_1/\lambda_2)}{\lambda_1 - \lambda_2}}$.

(f) With $\lambda_1 = \ln 2/(3.10 \text{ min})$, $\lambda_2 = \ln 2/(26.8 \text{ min})$, this formula gives

$$t_m = \frac{\ln(\lambda_1/\lambda_2)}{\lambda_1 - \lambda_2}$$

$$= \frac{\ln\left[\frac{\ln 2/(3.10 \text{ min})}{\ln 2/(26.8 \text{ min})}\right]}{\left(\frac{\ln 2}{3.10 \text{ min}} - \frac{\ln 2}{26.8 \text{ min}}\right)} = \frac{\ln\left(\frac{26.8 \text{ min}}{3.10 \text{ min}}\right)}{\ln 2\left(\frac{1}{3.10 \text{ min}} - \frac{1}{26.8 \text{ min}}\right)}$$

$$= \boxed{10.9 \text{ min}}$$

This result is in agreement with the result of part (d).

Section 44.5 The Decay Processes

P44.35 Atomic masses are given in Table 44.2.

(a) For this e^+ decay,

$$Q = (M_X - M_Y - 2m_e)c^2$$

$$= [39.962\,591 \text{ u} - 39.963\,999 \text{ u} - 2(0.000\,549 \text{ u})] \times (931.5 \text{ MeV/u})$$

$$Q = -2.33 \text{ MeV}$$

Since $Q < 0$, the decay $\boxed{\text{cannot occur}}$ spontaneously.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(b) For this alpha decay,

$$Q = \left(M_X - M_Y - 2m_e\right)c^2$$
$$= \left[97.905\,287\ \text{u} - 4.002\,603\ \text{u} - 93.905\,088\ \text{u}\right] \times \left(931.5\ \text{MeV/u}\right)$$
$$Q = -2.24\ \text{MeV}$$

Since $Q < 0$, the decay $\boxed{\text{cannot occur}}$ spontaneously.

(c) For this alpha decay,

$$Q = \left(M_X - M_Y - 2m_e\right)c^2$$
$$= \left[143.910\,083\ \text{u} - 4.002\,603\ \text{u} - 139.905\,434\ \text{u}\right] \times \left(931.5\ \text{MeV/u}\right)$$
$$Q = 1.91\ \text{MeV}$$

Since $Q > 0$, the decay $\boxed{\text{can occur}}$ spontaneously.

P44.36 (a) The reaction is $\quad {}_1^3\text{H} \rightarrow {}_2^3\text{He} + \text{e}^- + \bar{\nu}$.

Adding one electron, the reaction becomes

$${}_1^3\text{H nucleus} + \text{e}^- \rightarrow {}_2^3\text{He nucleus} + 2\text{e}^- + \bar{\nu}$$

Ignoring the slight difference in ionization energies, we have

$${}_1^3\text{H atom} \rightarrow {}_2^3\text{He atom} + \bar{\nu}$$

(b) The total energy released is the Q value:

$$Q = \left(M_{\text{H-3}} - M_{\text{He-3}}\right)c^2$$
$$Q = \left(3.016\,049\ \text{u} - 3.016\,029\ \text{u}\right)\left(931.5\ \text{MeV/u}\right)$$
$$= 0.018\,6\ \text{MeV} = \boxed{18.6\ \text{keV}}$$

P44.37 From Equation 44.21, carbon-14 undergoes beta decay:

$${}_6^{14}\text{C} \rightarrow {}_7^{14}\text{N} + \text{e}^- + \bar{\nu}$$

Adding six electrons to each side, this is the same as

$${}_6^{14}\text{C atom} \rightarrow {}_7^{14}\text{N atom} + \bar{\nu}$$

The Q value is

$$Q = \left(M_{\text{C-14}} - M_{\text{N-14}} - m_\nu\right)c^2$$
$$= \left[14.003\,242\ \text{u} - 14.003\,074\ \text{u} - 0\right]\left(931.5\ \text{MeV/u}\right)$$
$$= \boxed{0.156\ \text{MeV}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P44.38 Total Z and A are conserved.

(a) A gamma ray has zero charge and it contains no protons or neutrons. So for a gamma ray $Z = 0$ and $A = 0$. Keeping the total values of Z and A for the system conserved requires $Z = 28$ and $A = 65$ for X. With this atomic number it must be nickel, and the nucleus must be in an excited state, so X is $\boxed{{}^{65}_{28}\text{Ni}^{*}}$.

(b) An alpha particle, $\alpha = {}^{4}_{2}\text{He}$, has $Z = 2$ and $A = 4$. Total initial Z is 84, and total initial A is 215, so for X we require

$$Z = 84 = Z_X + 2 \quad \rightarrow \quad Z_X = 82 \quad \rightarrow \quad \text{Pb, and}$$

$$A = 215 = A_X + 4 \quad \rightarrow \quad A_X = 211, \quad \rightarrow \quad \text{X is } \boxed{{}^{211}_{82}\text{Pb}}.$$

(c) A positron, $e^{+} = {}^{0}_{1}e$, has charge the same as a nucleus with $Z = 1$. A neutrino, ${}^{0}_{0}\nu$, has no charge. Neither contains any protons or neutrons. So X must have by conservation $Z = 26 + 1 + 0 = 27$; so, X is Co. And $A = 55 + 0 + 0 = 55$: X is $\boxed{{}^{55}_{27}\text{Co}}$.

P44.39 Atomic masses are given in Table 44.2. We calculate the energy released by the reaction, its Q-value, as

$$Q = \left(M_{\text{U-238}} - M_{\text{Th-234}} - M_{\text{He-4}}\right)c^2$$

$$Q = (238.050\ 783 - 234.043\ 596 - 4.002\ 603)\ \text{u}(931.5\ \text{MeV/u})$$

$$= \boxed{4.27\ \text{MeV}}$$

P44.40 (a) The decay constant is $\lambda = \ln 2/10\ \text{h} = 0.0693/\text{h}$. The number of parent nuclei is given by $N_P = N_{P,0}e^{-\lambda t} = \left(1.00 \times 10^6\right)e^{-0.0693t}$, where t is in hours.

The number of daughter nuclei is equal to the number of missing parent nuclei,

$$\boxed{N_d = N_{P,0} - N_{P,0}e^{-\lambda t} = \left(1.00 \times 10^6\right)\left(1 - e^{-0.0693t}\right), \text{ where } t \text{ is in hours.}}$$

(b) The number of daughter nuclei starts from zero at $t = 0$. The number of stable product nuclei always increases with time and asymptotically approaches 1.00×10^6 as t increases without limit.

(c) The minimum number of daughter nuclei is zero at $t = 0$. The maximum number of daughter nuclei asymptotically approaches 1.00×10^6 as t increases without limit.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(d) The rate of change is

$$\frac{dN_d}{dt} = \left(1.00\times10^{6}\right)\left(0 + 0.0693\ e^{-0.0693t}\right) = 6.93\times10^{4}\,e^{-0.0693t}$$

where $\frac{dN_d}{dt}$ is in decays per hour and t is in hours. The rate of change has its maximum value, $6.93 \times 10^4\ \text{h}^{-1}$, at $t = 0$, after which the rate decreases more and more, approaching zero as t increases without limit.

P44.41 (a) The reaction for one particle is $\boxed{e^- + p \rightarrow n + \nu}$.

(b) For nuclei, ${}^{15}_{8}\text{O} + e^- \rightarrow {}^{15}_{7}\text{N} + \nu$.

Add seven electrons to both sides to obtain

$${}^{15}_{8}\text{O atom} \rightarrow {}^{15}_{7}\text{N atom} + \nu$$

From Table 44.2 of atomic masses,

$$Q = (15.003\ 065\ \text{u} - 15.000\ 109\ \text{u})(931.5\ \text{MeV/u})$$

$$= \boxed{2.75\ \text{MeV}}$$

P44.42 (a) The number of carbon atoms in the sample is

$$N_C = \left(\frac{0.021\ 0\ \text{g}}{12.0\ \text{g/mol}}\right)\left(\frac{6.02\times10^{23}\ \text{atoms}}{\text{mol}}\right) = \boxed{1.05\times10^{21}}$$

(b) 1 in 7.70×10^{11} carbon atoms is a ${}^{14}\text{C}$ atom. Then,

$$(N_0)_{\text{C-14}} = 1.05\times10^{21}\left(\frac{1}{7.70\times10^{11}}\right) = \boxed{1.37\times10^{9}}$$

(c) The decay constant for ${}^{14}\text{C}$ is

$$\lambda_{\text{C-14}} = \frac{\ln 2}{5\,730\ \text{yr}} = 1.21\times10^{-4}\ \text{yr}^{-1}\left(\frac{1\ \text{yr}}{3.16\times10^{7}\ \text{s}}\right)$$

$$= \boxed{3.83\times10^{-12}\ \text{s}^{-1}}$$

(d) We use $R = \lambda N = \lambda N_0 e^{-\lambda t}$. At $t = 0$,

$$R_0 = \lambda N_0 = \left(3.83\times10^{-12}\ \text{s}^{-1}\right)\left(1.37\times10^{9}\right)\left[\frac{7(86\ 400\ \text{s})}{1\ \text{week}}\right]$$

$$= \boxed{3.17\times10^{3}\ \text{decays/week}}$$

(e) At time t, $R = \frac{837}{0.880} = \boxed{951\ \text{decays/week}}$.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(f) Taking logarithms,

$$\ln\frac{R}{R_0} = -\lambda t \quad \text{so} \quad t = \frac{-1}{\lambda}\ln\left(\frac{R}{R_0}\right)$$

and

$$t = \frac{-1}{1.21\times10^{-4}\ \text{yr}^{-1}}\ln\left(\frac{951}{3.17\times10^{3}}\right) = \boxed{9.95\times10^{3}\ \text{yr}}$$

Section 44.6 Natural Radioactivity

P44.43 (a) The conversion is

$$4.00\ \text{pCi/L} = \left(\frac{4.00\times10^{-12}\ \text{Ci}}{1\ \text{L}}\right)\left(\frac{3.70\times10^{10}\ \text{Bq}}{1\ \text{Ci}}\right)\left(\frac{1.00\times10^{3}\ \text{L}}{1\ \text{m}^3}\right)$$

$$= \boxed{148\ \text{Bq/m}^3}$$

(b) Each cubic meter of air contains

$$N = \frac{R}{\lambda} = R\left(\frac{T_{1/2}}{\ln 2}\right) = \left(148\ \text{Bq/m}^3\right)\left(\frac{3.82\ \text{d}}{\ln 2}\right)\left(\frac{86\,400\ \text{s}}{1\ \text{d}}\right)$$

$$= \boxed{7.05\times10^{7}\ \text{atoms/m}^3}$$

(c) The density of radon in each cubic meter of air is

$$\text{density} = \left(7.05\times10^{7}\ \text{atoms/m}^3\right)\left(\frac{1\ \text{mol}}{6.02\times10^{23}\ \text{atoms}}\right)\left(\frac{222\ \text{g}}{1\ \text{mol}}\right)$$

$$= 2.60\times10^{-14}\ \text{g/m}^3$$

Since air has a density of 1.20 kg/m^3, the fraction consisting of radon is

$$\text{fraction} = \frac{2.60\times10^{-14}\ \text{g/m}^3}{1\,200\ \text{g/m}^3} = \boxed{2.17\times10^{-17}}$$

P44.44 The number of radon atoms remaining is

$$N = N_0 e^{-(\ln 2)t/T_{1/2}}$$

And the fraction remaining is

$$\frac{N}{N_0} = e^{-\lambda t} = e^{-(\ln 2)t/T_{1/2}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(a) With $T_{1/2} = 3.82$ d and $t = 7.00$ d,

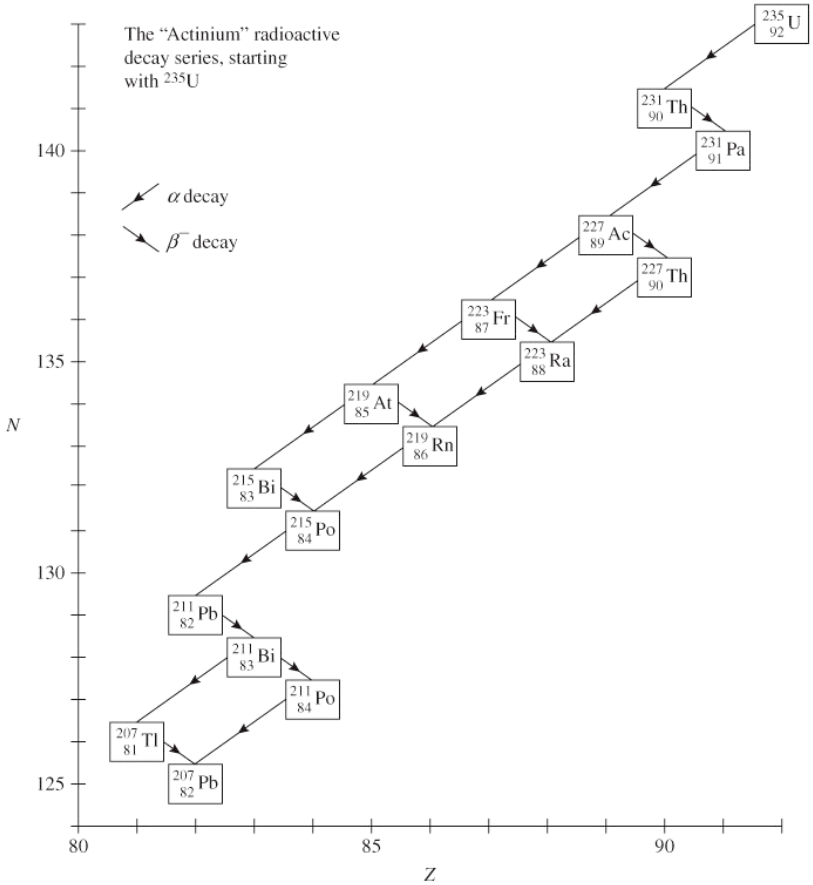
$$\frac{N}{N_0} = e^{-(\ln 2)(7.00)/(3.82)} = \boxed{0.281}$$

(b) When $t = 1.00$ yr $= 365.25$ d,

$$\frac{N}{N_0} = e^{-(\ln 2)(365.25)/(3.82)} = \boxed{1.65 \times 10^{-29}}$$

(c) Radon is continuously created as one daughter in the series of decays starting from the long-lived isotope ^{238}U.

P44.45 We find the chemical name by looking up Z in a periodic table. The values in the shaded boxes (^{235}U and ^{207}Pb) in Figure P44.45 were given; all others have been filled in as part of the solution shown in ANS. FIG. P44.45 below.



ANS. FIG. P44.45

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P44.46 (a) Let N be the number of ^{238}U nuclei and N' be ^{206}Pb nuclei.

Then $N = N_0 e^{-\lambda t}$ and $N_0 = N + N'$ so $N = (N + N')e^{-\lambda t}$ or $e^{\lambda t} = 1 + \dfrac{N'}{N}$. Taking logarithms,

$$\lambda t = \ln\left(1 + \frac{N'}{N}\right) \qquad \text{where} \qquad \lambda = \frac{\ln 2}{T_{1/2}}$$

Thus,

$$t = \left(\frac{T_{1/2}}{\ln 2}\right)\ln\left(1 + \frac{N'}{N}\right)$$

If $\dfrac{N}{N'} = 1.164$ for the $^{238}\text{U} \rightarrow {}^{206}\text{Pb}$ chain with $T_{1/2} = 4.47 \times 10^9$ yr, the age is:

$$t = \left(\frac{4.47 \times 10^9 \text{ yr}}{\ln 2}\right)\ln\left(1 + \frac{1}{1.164}\right) = \boxed{4.00 \times 10^9 \text{ yr}}$$

(b) From above, $e^{\lambda t} = 1 + \dfrac{N'}{N}$. Solving for $\dfrac{N}{N'}$ gives $\dfrac{N}{N'} = \dfrac{e^{-\lambda t}}{1 - e^{-\lambda t}}$.

With $T = 4.00 \times 10^9$ yr and $T_{1/2} = 7.04 \times 10^8$ yr for the $^{235}\text{U} \rightarrow {}^{207}\text{Pb}$ chain,

$$\lambda t = \left(\frac{\ln 2}{T_{1/2}}\right)t = \frac{(\ln 2)(4.00 \times 10^9 \text{ yr})}{7.04 \times 10^8 \text{ yr}} = 3.938$$

and $\boxed{\dfrac{N}{N'} = 0.019\,9 \text{ for the } {}^{235}\text{U to } {}^{207}\text{Pb chain.}}$

With $T = 4.00 \times 10^9$ yr and $T_{1/2} = 1.41 \times 10^{10}$ yr for the $^{232}\text{Th} \rightarrow {}^{208}\text{Pb}$ chain,

$$\lambda t = \frac{(\ln 2)(4.00 \times 10^9 \text{ yr})}{1.41 \times 10^{10} \text{ yr}} = 0.196\,6$$

and $\boxed{\dfrac{N}{N'} = 4.60 \text{ for the } {}^{232}\text{Th to } {}^{208}\text{Pb chain.}}$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Section 44.7 Nuclear Reactions

P44.47 Neglect recoil of product nucleus (i.e., do not require momentum conservation for the system of colliding particles). The energy balance gives $K_{\text{emerging}} = K_{\text{incident}} + Q$. To find Q,

$$Q = \left[\left(M_{\text{H}} + M_{\text{Al}}\right) - \left(M_{\text{Si}} + m_n\right)\right]c^2$$

$$Q = \left[(1.007\ 825 + 26.981\ 539) - (26.986\ 705 + 1.008\ 665)\right]\ \text{u} \times (931.5\ \text{MeV/u})$$

$$= -5.59\ \text{MeV}$$

Thus, $K_{\text{emerging}} = 6.61\ \text{MeV} - 5.59\ \text{MeV} = \boxed{1.02\ \text{MeV}}$.

P44.48 (a) The Q value of the reaction is given by

$$Q = \left[M_{^9\text{Be}} + M_{^4\text{He}} - M_{^{12}\text{C}} - m_n\right]c^2$$

$$Q = [9.012\ 182\ \text{u} + 4.002\ 603\ \text{u} - 12.000\ 000\ \text{u} - 1.008\ 665\ \text{u}] \times (931.5\ \text{MeV/u})$$

$$= \boxed{5.70\ \text{MeV}}$$

(b) For this reaction,

$$Q = \left[2M_{^2\text{H}} - M_{^3\text{He}} - m_n\right]$$

$$Q = [2(2.014\ 102\ \text{u}) - 3.016\ 029\ \text{u} - 1.008\ 665\ \text{u}] \times (931.5\ \text{MeV/u})$$

$$= \boxed{3.27\ \text{MeV}}$$

(c) The reaction in part (b) is exothermic because the Q value is positive.

P44.49 Total A and total Z are conserved.

(a) For X, $A = 24 + 1 - 4 = 21$ and $Z = 12 + 0 - 2 = 10$, so X is $\boxed{^{21}_{10}\text{Ne}}$.

(b) $A = 235 + 1 - 90 - 2 = 144$ and $Z = 92 + 0 - 38 - 0 = 54$, so X is $\boxed{^{144}_{54}\text{Xe}}$.

(c) $A = 2 - 2 = 0$ and $Z = 2 - 1 = +1$, so X must be a positron.

As it is ejected, it is accompanied by a neutrino:

$X + X' = \boxed{^0_1\text{e}^+ + \nu}$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P44.50 (a) $\boxed{{}^{197}_{79}\text{Au} + {}^{1}_{0}\text{n} \rightarrow {}^{198}_{79}\text{Au}^{*} \rightarrow {}^{198}_{80}\text{Hg} + {}^{0}_{-1}\text{e} + \bar{\nu}}$

Note the conservation of baryon number (which you can think of as nucleon census number and call mass number in this chapter) in the superscripts: 197 + 1 = 198 + 0. Note the conservation of charge in the subscripts: 79 + 0 = 80 – 1.

(b) Consider adding 79 electrons:

$${}^{197}_{79}\text{Au atom} + {}^{1}_{0}\text{n} \rightarrow {}^{198}_{80}\text{Hg atom} + \bar{\nu} + Q$$

Then,

$$Q = \left[M_{{}^{197}\text{Au}} + m_n - M_{{}^{198}\text{Hg}} \right] c^2$$

$$Q = [196.966\ 552\ \text{u} + 1.008\ 665\ \text{u} - 197.966\ 752\ \text{u}] \times (931.5\ \text{MeV/u})$$

$$= \boxed{7.89\ \text{MeV}}$$

P44.51 We consult Table 44.2 for the masses. For the first reaction,

$${}^{9}_{4}\text{Be} + 1.665\ \text{MeV} \rightarrow {}^{8}_{4}\text{Be} + {}^{1}_{0}\text{n}$$

so $$M_{{}^{8}_{4}\text{Be}} = M_{{}^{9}_{4}\text{Be}} - \frac{Q}{c^2} - m_n$$

$$M_{{}^{8}_{4}\text{Be}} = 9.012\,182\ \text{u} - \frac{(-1.665\ \text{MeV})}{931.5\ \text{MeV/u}} - 1.008\,665\ \text{u}$$

$$= \boxed{8.005\,3\ \text{u}}$$

For the second reaction,

$${}^{9}_{4}\text{Be} + {}^{1}_{0}\text{n} \rightarrow {}^{10}_{4}\text{Be} + 6.812\ \text{MeV}$$

so $$M_{{}^{10}_{4}\text{Be}} = M_{{}^{9}_{4}\text{Be}} + m_n - \frac{Q}{c^2}$$

$$M_{{}^{10}_{4}\text{Be}} = 9.012\,182\ \text{u} + 1.008\,665\ \text{u} - \frac{6.812\ \text{MeV}}{931.5\ \text{MeV/u}}$$

$$= \boxed{10.013\,5\ \text{u}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Section 44.8 Nuclear Magnetic Resonance and Magnetic Resonance Imaging

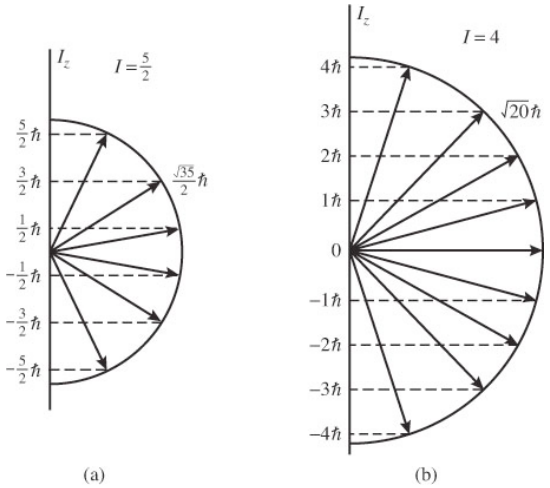
P44.52 It is the quantum particle under boundary conditions model that is behind the general rules: With angular momentum quantum number I, the magnitude of the angular momentum must be $\sqrt{I(I+1)}\ \hbar$. Whether I is an integer or a half-integer, the allowed values for one component of angular momentum being measured range from $+I\hbar$ to $+(I-1)\hbar$ to … to $-I\hbar$. Conditions that the wave function for a quantum particle must satisfy, for self-consistency under rotations in three-dimensional space, impose these requirements. We call a component being measured the z component. It can be measured more directly, as in a nuclear magnetic resonance experiment, or less directly, as from the way the angular momentum influences the intrinsic energy levels of a system and the number of available states within an energy level.

(a) With $I = 5/2$, the magnitude of the angular momentum is

$$\begin{aligned}\sqrt{I(I+1)}\ \hbar &= \sqrt{\tfrac{5}{2}(\tfrac{5}{2}+1)}\ \hbar = \sqrt{35}\ \hbar/2 \\ &= 2.958\ 04(6.626\times10^{-34}\ \text{J}\cdot\text{s})/2\pi \\ &= 3.119\times10^{-34}\ \text{kg}\cdot\text{m}^2/\text{s}\end{aligned}$$

The z component can take the values $+5\hbar/2$, $+3\hbar/2$, $+\hbar/2$, $-\hbar/2$, $-3\hbar/2$, and $-5\hbar/2$. These identifications are shown in ANS. FIG. P44.52(a).

(b) Similarly, with $I = 4$, the magnitude of the angular momentum of a nucleus is $\sqrt{I(I+1)}\ \hbar = \sqrt{4(4+1)}\ \hbar = \sqrt{20}\ \hbar$ and its z component must have one of the nine values $+4\hbar$, $+3\hbar$, $+2\hbar$, $+\hbar$, 0, $-\hbar$, $-2\hbar$, $-3\hbar$, $-4\hbar$, as shown in ANS. FIG. P44.52(b).



ANS. FIG. P44.52

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P44.53 From page 1406 and Equation 44.31, the magnetic moment is $-1.913\,5\mu_n$ for the neutron and $2.792\,8\mu_n$ for the proton, where $\mu_n = 5.05\times10^{-27}\ \text{J/T}$ is the nuclear magneton.

(a) The Larmor frequency of free neutrons is

$$f_n = \frac{|2\mu B|}{h} = \frac{2\left[(1.913\,5)(5.05\times10^{-27}\ \text{J/T})\right](1.00\ \text{T})}{6.626\times10^{-34}\ \text{J}\cdot\text{s}}$$

$$= \boxed{29.2\ \text{MHz}}$$

(b) The Larmor frequency of free protons is

$$f_p = \frac{2\left[(2.792\,8)(5.05\times10^{-27}\ \text{J/T})\right](1.00\ \text{T})}{6.626\times10^{-34}\ \text{J}\cdot\text{s}} = \boxed{42.6\ \text{MHz}}$$

(c) In the Earth's magnetic field,

$$f_p = \frac{2\left[(2.792\,8)(5.05\times10^{-27}\ \text{J/T})\right](50.0\times10^{-6}\ \text{T})}{6.626\times10^{-34}\ \text{J}\cdot\text{s}} = \boxed{2.13\ \text{kHz}}$$

Additional Problems

***P44.54** From $R = R_0 e^{-\lambda t}$ and $T_{1/2} = 5\,730$ yr for ^{14}C, the age of the sample is

$$t = \frac{-1}{\lambda}\ln\left(\frac{R}{R_0}\right) = -T_{1/2}\left[\frac{\ln(R/R_0)}{\ln 2}\right] = -(5\,730\ \text{yr})\left[\frac{\ln(0.600)}{\ln 2}\right]$$

$$= \boxed{4.22\times10^{3}\ \text{yr}}$$

***P44.55** From $R = R_0 e^{-\lambda t}$, the elapsed time is

$$t = \frac{-1}{\lambda}\ln\left(\frac{R}{R_0}\right) = -T_{1/2}\left[\frac{\ln(R/R_0)}{\ln 2}\right] = -(14.0\ \text{d})\left[\frac{\ln\left(\frac{20.0\ \text{mCi}}{200\ \text{mCi}}\right)}{\ln 2}\right]$$

$$= \boxed{46.5\ \text{d}}$$

P44.56 The proposed reaction can be written as

$$^{10}_{5}\text{B} + {}^{4}_{2}\text{He} \quad \rightarrow \quad {}^{1}_{1}\text{H} + {}^{12}_{6}\text{C}$$

While electric charge is conserved (5 + 2 = 1 + 6), the number of nucleons is not ($10 + 4 \neq 1 + 12$). Therefore, this reaction cannot occur.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P44.57 (a) From Equation 44.1,

$$r = aA^{1/3} = (1.2\text{ fm})A^{1/3} = (1.2\times10^{-15}\text{ m})A^{1/3}$$

When $A = 12$, $r = 1.2\text{ fm}(12)^{1/3} = 2.7\times10^{-15}\text{ m} = \boxed{2.7\text{ fm}}$

(b) $$F = \frac{k_e(Z-1)e^2}{r^2} = \frac{(8.99\times10^9\text{ N}\cdot\text{m}^2/\text{C}^2)(Z-1)(1.60\times10^{-19}\text{ C})^2}{r^2}$$

When $Z = 6$ and $r = (1.2\times10^{-15}\text{ m})(12)^{1/3}$, $F = \boxed{1.5\times10^2\text{ N}}$.

(c) $$U = \frac{k_e q_1 q_2}{r} = \frac{k_e(Z-1)e^2}{r} = \frac{(8.99\times10^9)(Z-1)(1.6\times10^{-19})^2}{r}$$

When $Z = 6$ and $r = (1.2\times10^{-15}\text{ m})(12)^{1/3}$,

$U = 4.2\times10^{-13}\text{ J} = \boxed{2.6\text{ MeV}}$

(d) $A = 238$, $Z = 92$, and $r = 1.2\text{ fm}(238)^{1/3} = 7.4\times10^{-15}\text{ m} = \boxed{7.4\text{ fm}}$

$F = \boxed{3.8\times10^2\text{ N}}$ and $U = 2.8\times10^{-12}\text{ J} = \boxed{18\text{ MeV}}$

P44.58 (a) The process cannot occur because energy input would be required. Note that the mass of the proton is less than the sum of the masses of the neutron and positron (electron):

$$m_n + m_{e^+} > m_p$$
$$1.008\,665\text{ u} + 0.000\,549\text{ u}$$
$$1.009\,214\text{ u} > 1.007\,276\text{ u}$$

Therefore, the reaction $\text{p} \rightarrow \text{n} + \text{e}^+ + \nu$ would violate the law of conservation of energy.

(b) The required energy can come from the electrostatic repulsion of protons in the parent nucleus.

(c) Add seven electrons to both sides of the reaction for nuclei ${}^{13}_{7}\text{N} \rightarrow {}^{13}_{6}\text{C} + \text{e}^+ + \nu$ to obtain the reaction for neutral atoms ${}^{13}_{7}\text{N atom} \rightarrow {}^{13}_{6}\text{C atom} + \text{e}^+ + \text{e}^- + \nu$.

$$Q = [m({}^{13}\text{N}) - m({}^{13}\text{C}) - m_{e^+} - m_{e^-} - m_\nu]c^2$$
$$Q = [13.005\,739\text{ u} - 13.003\,355\text{ u} - 2(5.49\times10^{-4}\text{ u}) - 0] \times (931.5\text{ MeV/u})$$
$$= \boxed{1.20\text{ MeV}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P44.59 $E = -\vec{\boldsymbol{\mu}} \cdot \vec{\mathbf{B}}$ so the energies are $E_1 = +\mu B$ and $E_2 = -\mu B$, where B = 12.5 T, $\mu = 2.792\,8\mu_n$, and $\mu_n = 5.05 \times 10^{-27}$ J/T. The energy difference is

$$\Delta E = 2\mu B = 2(2.792\,8)(5.05 \times 10^{-27}\ \text{J/T})(12.5\ \text{T})$$
$$= 3.53 \times 10^{-25}\ \text{J} = 2.20 \times 10^{-6}\ \text{eV} = \boxed{2.20\ \mu\text{eV}}$$

P44.60 We check the Q value of this reaction:

$$Q = [238.050\,788\ \text{u} - 237.051\,144\ \text{u} - 1.007\,825\ \text{u}] \times (931.5\ \text{MeV/u})$$
$$= -7.62\ \text{MeV}$$

The Q value of this hypothetical decay is calculated to be –7.62 MeV, which means you would have to add this much energy to the ^{238}U nucleus to make it emit a proton.

P44.61 (a) The system of a separated proton and electron puts out energy 13.606 eV to become a hydrogen atom in its ground state. This decrease in its rest energy appears also as a decrease in mass: the mass is smaller.

(b) The mass difference is

$$|\Delta m| = \frac{|E|}{c^2} = \left[\frac{13.6\ \text{eV}}{(3.00 \times 10^8\ \text{m/s})^2}\right]\left(\frac{1.60 \times 10^{-19}\ \text{J}}{1\ \text{eV}}\right)$$
$$= (2.42 \times 10^{-35}\ \text{kg})\left(\frac{1\ \text{u}}{1.66 \times 10^{-27}\ \text{kg}}\right) = \boxed{1.46 \times 10^{-8}\ \text{u}}$$

(c) As a percentage of the total mass,

$$\frac{1.46 \times 10^{-8}\ \text{u}}{1.007\,825\ \text{u}} = 1.45 \times 10^{-8} = \boxed{1.45 \times 10^{-6}\%}$$

(d) No. The textbook table lists 1.007 825 u as the atomic mass of hydrogen. This correction of 0.000 000 01 u is on the order of 100 times too small to affect the values listed.

P44.62 We check the Q value of the ^{57}Co nuclei decay by e^+:

$$^{57}_{27}\text{Co} \rightarrow {}^{57}_{26}\text{Fe} + {}^{0}_{+1}\text{e} + \bar{\nu}$$

Mass values appear in Table 44.2. For this reaction,

$$Q = [56.936\,291 - 56.935\,394 - 2(0.000\,549)]\text{u}(931.5\ \text{MeV/u})$$
$$= -0.187\ \text{MeV}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

The nucleus ^{57}Co cannot decay by e^+ emission because the Q value is -0.187 MeV.

P44.63 (a) The number of nuclei at $t = 0$ is given by

$$N_0 = \frac{\text{mass}}{\text{mass per atom}} = \frac{1.00 \text{ kg}}{(239.05 \text{ u})\left(1.66 \times 10^{-27} \text{ kg/u}\right)}$$

$$= \boxed{2.52 \times 10^{24}}$$

(b) To find the initial activity, we first compute the ecay constant:

$$\lambda = \frac{\ln 2}{T_{1/2}} = \frac{\ln 2}{\left(2.412 \times 10^4 \text{ yr}\right)\left(3.156 \times 10^7 \text{ s/yr}\right)} = 9.106 \times 10^{-13} \text{ s}^{-1}$$

Then,

$$R_0 = \lambda N_0 = \left(9.106 \times 10^{-13} \text{ s}^{-1}\right)\left(2.52 \times 10^{24}\right) = \boxed{2.29 \times 10^{12} \text{ Bq}}$$

(c) From $R = R_0 e^{-\lambda t}$,

$$t = \frac{-1}{\lambda} \ln\left(\frac{R}{R_0}\right) = \frac{1}{\lambda} \ln\left(\frac{R_0}{R}\right)$$

$$t = \frac{1}{9.106 \times 10^{-13} \text{ s}^{-1}} \ln\left(\frac{2.29 \times 10^{12} \text{ Bq}}{0.100 \text{ Bq}}\right)$$

$$= 3.38 \times 10^{13} \text{ s}\left(\frac{1 \text{ yr}}{3.156 \times 10^7 \text{ s}}\right) = \boxed{1.07 \times 10^6 \text{ yr}}$$

P44.64 (a) One liter of milk contains this many ^{40}K nuclei:

$$N = (2.00 \text{ g})\left(\frac{6.02 \times 10^{23} \text{ nuclei/mol}}{39.1 \text{ g/mol}}\right)\left(\frac{0.011\,7}{100}\right)$$

$$= 3.60 \times 10^{18} \text{ nuclei}$$

$$\lambda = \frac{\ln 2}{T_{1/2}} = \frac{\ln 2}{1.28 \times 10^9 \text{ yr}}\left(\frac{1 \text{ yr}}{3.156 \times 10^7 \text{ s}}\right) = 1.72 \times 10^{-17} \text{ s}^{-1}$$

$$R = \lambda N = \left(1.72 \times 10^{-17} \text{ s}^{-1}\right)\left(3.60 \times 10^{18}\right) = 61.8 \text{ Bq}$$

The activity is $\boxed{61.8 \text{ Bq/L}}$.

(b) For the iodine, $R = R_0 e^{-\lambda t}$, with $\lambda = \frac{\ln 2}{8.04 \text{ d}}$. Then,

$$t = \frac{1}{\lambda} \ln\left(\frac{R_0}{R}\right) = \frac{8.04 \text{ d}}{\ln 2} \ln\left(\frac{2\,000}{61.8}\right) = \boxed{40.3 \text{ d}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P44.65 We have $N_{\text{U-235}} = N_{0,\text{ U-235}} e^{-\lambda_{\text{U-235}} t}$

and $N_{\text{U-238}} = N_{0,\text{ U-238}} e^{-\lambda_{\text{U-238}} t}$,

so $$\frac{N_{\text{U-235}}}{N_{\text{U-238}}} = 0.007\ 25 = e^{\left(-(\ln 2)t/T_{1/2,\text{ U-235}} + (\ln 2)t/T_{1/2,\text{ U-238}}\right)}$$

Taking logarithms,

$$-4.93 = \left(-\frac{\ln 2}{0.704 \times 10^9 \text{ yr}} + \frac{\ln 2}{4.47 \times 10^9 \text{ yr}} \right) t$$

$$-4.93 = \left(-\frac{1}{0.704 \times 10^9 \text{ yr}} + \frac{1}{4.47 \times 10^9 \text{ yr}} \right) (\ln 2) t$$

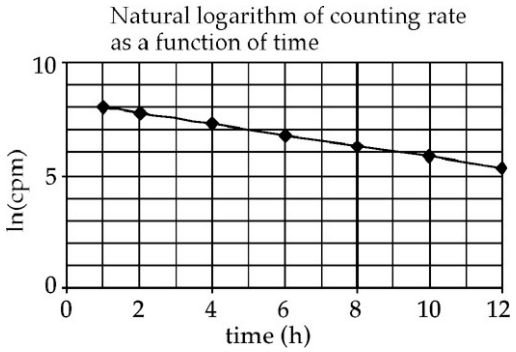
$$t = \frac{-4.93}{\left(-1.20 \times 10^{-9} \text{ yr}^{-1}\right) \ln 2} = 5.94 \times 10^9 \text{ yr} = \boxed{5.94 \text{ Gyr}}$$

P44.66 (a) See ANS. FIG. P44.66. A least-square fit to the graph yields:

$$\lambda = -\text{slope} = -\left(-0.250 \text{ h}^{-1}\right) = 0.250 \text{ h}^{-1}$$

and

$$\ln(\text{cpm})\big|_{t=0} = \text{intercept} = 8.30$$



ANS. FIG. P44.66

(b) From part (a),

$$\lambda = 0.250 \text{ h}^{-1} \left(\frac{1 \text{ h}}{60.0 \text{ min}} \right) = \boxed{4.17 \times 10^{-3} \text{ min}^{-1}}$$

and $$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{\ln 2}{4.17 \times 10^{-3} \text{ min}^{-1}} = 166 \text{ min} = \boxed{2.77 \text{ h}}$$

(c) From part (a), $\text{intercept} = \ln(\text{cpm})_0 = 8.30$.

Thus, $(\text{cpm})_0 = e^{8.30}$ counts/min $= \boxed{4.02 \times 10^3 \text{ counts/min}}$.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(d) At $t = 0$,

$$N_0 = \frac{R_0}{\lambda} = \frac{1}{\lambda}\frac{(\text{cpm})_0}{\text{Eff}} = \frac{4.02 \times 10^3 \text{ counts/min}}{(4.17 \times 10^{-3} \text{ min}^{-1})(0.100)}$$

$$= \boxed{9.65 \times 10^6 \text{ atoms}}$$

P44.67 (a) If ΔE is the energy difference between the excited and ground states of the nucleus of mass M, and hf is the energy of the emitted photon, conservation of energy for the nucleus-photon system gives

$$\Delta E = hf + E_r \quad \textbf{[1]}$$

where E_r is the recoil energy of the nucleus, which can be expressed as

$$E_r = \frac{Mv^2}{2} = \frac{(Mv)^2}{2M} \quad \textbf{[2]}$$

Since system momentum must also be conserved, we have

$$Mv = \frac{hf}{c} \quad \textbf{[3]}$$

Hence, E_r can be expressed as $E_r = \dfrac{(hf)^2}{2Mc^2}$.

When $hf << Mc^2$, we can make the approximation that $hf \approx \Delta E$,

so $E_r \approx \dfrac{(\Delta E)^2}{2Mc^2}$.

(b) $E_r = \dfrac{(\Delta E)^2}{2Mc^2}$, where $\Delta E = 0.014\ 4 \text{ MeV}$

and $Mc^2 = (57 \text{ u})(931.5 \text{ MeV/u}) = 5.31 \times 10^4 \text{ MeV}$.

Therefore,

$$E_r = \frac{(1.44 \times 10^{-2} \text{ MeV})^2}{2(5.31 \times 10^4 \text{ MeV})} = 1.95 \times 10^{-9} \text{ MeV} = \boxed{1.95 \times 10^{-3} \text{ eV}}$$

P44.68 (a) If we assume all the ^{87}Sr came from ^{87}Rb, then $N = N_0 e^{-\lambda t}$ yields

$$t = \frac{-1}{\lambda}\ln\left(\frac{N}{N_0}\right) = \frac{T_{1/2}}{\ln 2}\ln\left(\frac{N_0}{N}\right)$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

where $N = N_{\text{Rb-87}}$

and $N_0 = N_{\text{Sr-87}} + N_{\text{Rb-87}}$.

$$t = \frac{\left(4.75\times10^{10}\ \text{yr}\right)}{\ln 2}\ln\left(\frac{1.82\times10^{10}+1.07\times10^{9}}{1.82\times10^{10}}\right) = \boxed{3.91\times10^{9}\ \text{yr}}$$

(b) It could be $\boxed{\text{no older.}}$ The rock could be younger if some ^{87}Sr were originally present. We must make some assumption about the original quantity of radioactive material. In part (a) we assumed that the rock originally contained no strontium.

P44.69 The time of flight is given by $\Delta t = d/v$. Since $K = \frac{1}{2}mv^2$,

$$\Delta t = \frac{d}{\sqrt{\frac{2K}{m}}} = \frac{10.0\times10^{3}\ \text{m}}{\sqrt{\frac{2(0.040\ 0\ \text{eV})\left(1.60\times10^{-19}\ \text{J/eV}\right)}{1.67\times10^{-27}\ \text{kg}}}} = 3.61\ \text{s}$$

The decay constant is

$$\lambda = \frac{0.693}{T_{1/2}} = \frac{0.693}{(10.4\ \text{min})(60\ \text{s/min})} = 1.11\times10^{-3}\ \text{s}^{-1}$$

Therefore we have

$$\lambda\Delta t = \left(1.11\times10^{-3}\ \text{s}\right)(3.61\ \text{s}) = 4.01\times10^{-3} = 0.004\ 01$$

And the fraction remaining is

$$\frac{N}{N_0} = e^{-\lambda\Delta t} = e^{-0.004\ 01} = 0.996\ 0.$$

Hence, the fraction that has decayed in this time interval is

$$1 - \frac{N}{N_0} = 0.004\ 01 \quad \text{or} \quad \boxed{0.401\%}$$

P44.70 (a) For cobalt-56,

$$\lambda = \frac{\ln 2}{T_{1/2}} = \frac{\ln 2}{77.1\ \text{d}}\left(\frac{365.25\ \text{d}}{1\ \text{yr}}\right) = 3.28\ \text{yr}^{-1}$$

The elapsed time from July 1054 to July 2010 is 956 yr. Then, $R = R_0 e^{-\lambda t}$ implies

$$\frac{R}{R_0} = e^{-\lambda t} = e^{-\left(3.28\ \text{yr}^{-1}\right)(956\ \text{yr})} = e^{-3\,139} = e^{-(\ln 10)1\,363} = \boxed{\sim 10^{-1\,363}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(b) For carbon-14,

$$\lambda = \frac{\ln 2}{5\,730 \text{ yr}} = 1.21 \times 10^{-4} \text{ yr}^{-1}$$

and

$$\frac{R}{R_0} = e^{-\lambda t} = e^{-(1.21\times10^{-4}\ \text{yr}^{-1})(956\ \text{yr})} = e^{-0.116} = \boxed{0.891}$$

P44.71 (a) For the electron capture,

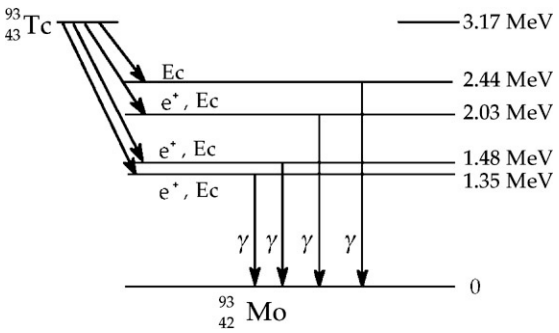
$${}^{93}_{43}\text{Tc} + {}^{0}_{-1}\text{e} \rightarrow {}^{93}_{42}\text{Mo} + \gamma$$

For positron emission,

$${}^{93}_{43}\text{Tc} \rightarrow {}^{93}_{42}\text{Mo} + {}^{0}_{+1}\text{e} + \gamma$$

The daughter nucleus in both forms of decay is $\boxed{{}^{93}_{42}\text{Mo}}$.

(b) We usually calculate the Q value under the assumption that the daughter nucleus is in its ground state, but for these decays, the Q value gives the upper limit of energy available to the daughter nucleus to be above its ground state.



ANS. FIG. P44.71

For electron capture, the disintegration energy is

$$Q = \left[M_{^{93}\text{Tc}} - M_{^{93}\text{Mo}}\right]c^2$$

$$Q = \left[92.910\,2\ \text{u} - 92.906\,8\ \text{u}\right](931.5\ \text{MeV/u})$$

$$= 3.17\ \text{MeV} > 2.44\ \text{MeV}$$

so $\boxed{\text{electron capture}}$ provides enough energy for ${}^{93}_{42}\text{Mo}$ to be in $\boxed{\text{all levels}}$ above its ground state.

For e^+ emission, the disintegration energy is

$$Q' = \left[M_{^{93}\text{Tc}} - M_{^{93}\text{Mo}} - 2m_e\right]c^2.$$

$$Q' = \left[92.910\ 2\ \text{u} - 92.906\ 8\ \text{u} - 2(0.000\ 549\ \text{u})\right](931.5\ \text{MeV/u})$$

$$= 2.14\ \text{MeV}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

so $\boxed{e^+ \text{ emission}}$ does not supply enough energy for $^{93}_{42}\text{Mo}$ to be in the 4.22 MeV state, $\boxed{\text{only 1.35 MeV, 1.48 MeV, and 1.35 MeV}}$ above ground (see ANS. FIG. P44.71).

P44.72 We start with $R = R_0 e^{-\lambda t}$, and take the natural logarithm of both sides, giving $\ln R = \ln R_0 - \lambda t$, which is the equation of a straight line with $|\text{slope}| = \lambda$. The logarithmic plot shown in Figure P44.72 is fitted by

$$\ln R = 8.44 - 0.262t$$

If t is measured in minutes, then decay constant λ is 0.262 per minute. The half–life is

$$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{\ln 2}{0.262/\text{min}} = \boxed{2.64 \text{ min}}$$

The reported half–life of ^{137}Ba is 2.55 min. The difference reflects experimental uncertainties.

P44.73 (a) With m_n and v_n as the mass and speed of the neutrons, Equation 9.24 for elastic collisions becomes for the two collisions, after making appropriate notational changes,

$$v_1 = \left(\frac{2m_n}{m_n + m_1}\right) v_n, \text{ and } v_2 = \left(\frac{2m_n}{m_n + m_2}\right) v_n$$

Solving,

$$(m_n + m_2)v_2 = (m_n + m_1)v_1 = 2m_n v_n$$

$$m_n(v_2 - v_1) = m_1 v_1 - m_2 v_2 \qquad \rightarrow \qquad m_n = \frac{m_1 v_1 - m_2 v_2}{v_2 - v_1}$$

(b) We obtain the neutron mass from

$$m_n = \frac{(1 \text{ u})(3.30 \times 10^7 \text{ m/s}) - (14 \text{ u})(4.70 \times 10^6 \text{ m/s})}{4.70 \times 10^6 \text{ m/s} - 3.30 \times 10^7 \text{ m/s}} = \boxed{1.16 \text{ u}}$$

P44.74 (a) We treat the collision of the two particles a and X as a perfectly inelastic collision: the kinetic energy that is converted into internal energy supplies the missing energy Q, permitting the conversion of the particles into Y and b.

Initially, the projectile M_a moves with velocity v_a while the target M_X is at rest. We have from momentum conservation for the projectile-target system:

$$M_a v_a = (M_a + M_X) v_c$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

The initial energy is

$$E_i = \frac{1}{2} M_{\text{a}} v_{\text{a}}^2$$

The final kinetic energy is:

$$E_f = \frac{1}{2}\left(M_{\text{a}} + M_{\text{X}}\right) v_c^2 = \frac{1}{2}\left(M_{\text{a}} + M_{\text{X}}\right)\left[\frac{M_{\text{a}} v_{\text{a}}}{M_{\text{a}} + M_{\text{X}}}\right]^2$$

$$= \left[\frac{M_{\text{a}}}{M_{\text{a}} + M_{\text{X}}}\right] E_i$$

From this, we see that E_f is always less than E_i and the change in energy, $E_f - E_i$, is given by

$$E_f - E_i = \left[\frac{M_{\text{a}}}{M_{\text{a}} + M_{\text{X}}} - 1\right] E_i = -\left[\frac{M_{\text{X}}}{M_{\text{a}} + M_{\text{X}}}\right] E_i$$

This loss of kinetic energy in the isolated system corresponds to an increase in mass-energy during the reaction. Thus, the absolute value of this kinetic energy change is equal to $-Q$ (remember that Q is negative in an endothermic reaction). The initial kinetic energy E_i is the threshold energy E_{th}. Therefore,

$$-Q = \left[\frac{M_{\text{X}}}{M_{\text{a}} + M_{\text{X}}}\right] E_{\text{th}}$$

or $$E_{\text{th}} = -Q\left[\frac{M_{\text{X}} + M_{\text{a}}}{M_{\text{X}}}\right] = -Q\left[1 + \frac{M_{\text{a}}}{M_{\text{X}}}\right].$$

(b) We first calculate the Q value for the reaction:

$$Q = \left[M_{\text{N-14}} + M_{\text{He-4}} - M_{\text{O-17}} - M_{\text{H-1}}\right] c^2$$

$$Q = \left[14.003\ 074\ \text{u} + 4.002\ 603\ \text{u} - 16.999\ 132\ \text{u} - 1.007\ 825\ \text{u}\right] \times \left(931.5\ \text{MeV/u}\right)$$

$$= -1.19\ \text{MeV}$$

Then,

$$E_{\text{th}} = -Q\left[\frac{M_{\text{X}} + M_{\text{a}}}{M_{\text{X}}}\right] = -(-1.19\ \text{MeV})\left[1 + \frac{4.002\ 603\ \text{u}}{14.003\ 074\ \text{u}}\right]$$

$$= \boxed{1.53\ \text{MeV}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P44.75 We have the following information: $N_X(0) = 2.50N_Y(0)$, $N_X(3\text{ d}) = 4.20N_Y(3\text{ d})$, and $T_{1/2Y} = 1.60$ d. The nuclei decay exponentially:

$$N_X(3\text{ d}) = 4.20N_Y(3\text{ d})$$

$$N_X(0)e^{-\lambda_X(3\text{ d})} = 4.20N_Y(0)e^{-\lambda_Y(3\text{ d})} = 4.20\frac{N_X(0)}{2.50}e^{-\lambda_Y(3\text{ d})}$$

$$e^{(3\text{ d})\lambda_X} = \frac{2.5}{4.2}e^{(3\text{ d})\lambda_Y}$$

Taking the natural logarithm of both sides,

$$(3\text{ d})\lambda_X = \ln\left(\frac{2.5}{4.2}\right) + (3\text{ d})\lambda_Y$$

$$(3\text{ d})\frac{0.693}{T_{1/2X}} = \ln\left(\frac{2.5}{4.2}\right) + (3\text{ d})\frac{0.693}{1.60\text{ d}} = 0.781$$

The half-life of X is $T_{1/2X} = \boxed{2.66\text{ d}}$

P44.76 We have the following information: $\frac{N_X(0)}{N_Y(0)} = r_1$, $\frac{N_X(\Delta t)}{N_Y(\Delta t)} = r_2$, and $T_{1/2Y} = T_Y$. The nuclei decay exponentially:

$$N_X(\Delta t) = r_2 N_Y(\Delta t)$$

$$N_X(0)e^{-\lambda_X \Delta t} = r_2 N_Y(0)e^{-\lambda_Y \Delta t} = \left(\frac{r_2}{r_1}\right)N_X(0)e^{-\lambda_Y \Delta t}$$

$$e^{-\Delta t\lambda_X} = \frac{r_2}{r_1}e^{-\Delta t\lambda_Y}$$

Taking the natural logarithm of both sides,

$$-\Delta t\lambda_X = \ln\left(\frac{r_2}{r_1}\right) - \Delta t\lambda_Y$$

$$\Delta t\frac{\ln 2}{T_X} = -\ln\left(\frac{r_2}{r_1}\right) + \Delta t\frac{\ln 2}{T_Y} = \ln\left(\frac{r_1}{r_2}\right) + \Delta t\frac{\ln 2}{T_Y}$$

$$\frac{1}{T_X} = \frac{\ln(r_1/r_2)}{\Delta t \ln 2} + \frac{1}{T_Y} = \frac{T_Y \ln(r_1/r_2) + \Delta t \ln 2}{T_Y \Delta t \ln 2} = \frac{\ln\left[2(r_1/r_2)^{T_Y/\Delta t}\right]}{T_Y \ln 2}$$

The half-life of X is $T_X = \boxed{\dfrac{T_Y \ln 2}{\ln\left[2(r_1/r_2)^{T_Y/\Delta t}\right]}}$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Challenge Problems

P44.77 The electric charge density in the sphere is

$$\rho = \frac{Ze}{(4/3)\pi R^3}$$

Using Gauss's Law inside the sphere,

$$E \cdot 4\pi r^2 = \left(\frac{(4/3)\pi r^3}{\epsilon_0} \right) \frac{Ze}{(4/3)\pi R^3} :$$

or $$E = \left(\frac{1}{4\pi \epsilon_0} \frac{Ze}{R^3} \right) r \qquad (r \le R)$$

Outside the sphere, the field is

$$E = \frac{1}{4\pi \epsilon_0} \frac{Ze}{r^2} \qquad (r \ge R)$$

We now find the electrostatic energy

$$U = \int_{r=0}^{\infty} \left(\frac{1}{2} \epsilon_0 E^2 \right) 4\pi r^2 \, dr$$

$$U = \frac{1}{2} \epsilon_0 \int_0^R \left[\left(\frac{Ze}{4\pi \epsilon_0 R^3} \right) r \right]^2 4\pi r^2 \, dr + \frac{1}{2} \epsilon_0 \int_R^{\infty} \left[\frac{1}{4\pi \epsilon_0} \frac{Ze}{r^2} \right]^2 4\pi r^2 \, dr$$

$$= 2\pi \epsilon_0 \left(\frac{Ze}{4\pi \epsilon_0} \right)^2 \int_0^R \left[\frac{r^2}{R^6} \right] r^2 \, dr + 2\pi \epsilon_0 \left(\frac{Ze}{4\pi \epsilon_0} \right)^2 \int_R^{\infty} \left[\frac{1}{r^4} \right] r^2 dr$$

$$= \frac{Z^2 e^2}{8\pi \epsilon_0} \left[\int_0^R \frac{r^4}{R^6} dr + \int_R^{\infty} \frac{dr}{r^2} \right] = \frac{Z^2 e^2}{8\pi \epsilon_0} \left[\left(\frac{R^5}{5R^6} \right) \Big|_0^R - \left(\frac{1}{r} \right) \Big|_R^{\infty} \right]$$

$$= \frac{Z^2 e^2}{8\pi \epsilon_0} \left[\frac{R^5}{5R^6} + \frac{1}{R} \right] = \frac{3}{20} \frac{Z^2 e^2}{\pi \epsilon_0 R} = \frac{3}{5} \left(\frac{1}{4\pi \epsilon_0} \right) \frac{Z^2 e^2}{R}$$

or $$\boxed{U = \frac{3}{20} \frac{Z^2 e^2}{\pi \epsilon_0 R} = \frac{3 k_e Z^2 e^2}{5R}}$$

P44.78 (a) Add two electrons to both sides of the reaction to have it in neutral-atom terms:

$$4\,{}_1^1\mathrm{H}\ \text{atom} \rightarrow {}_2^4\mathrm{He}\ \text{atom} + Q \ \rightarrow\ Q = \Delta m c^2 = \left[4M_{{}_1^1\mathrm{H}} - M_{{}_2^4\mathrm{He}} \right] c^2$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

The Q value is then

$$Q = \left[4(1.007\ 825\ \text{u}) - 4.002\ 603\ \text{u}\right] \times (931.5\ \text{MeV/u})\left(\frac{1.60\times10^{-13}\ \text{J}}{1\ \text{MeV}}\right)$$

$$= \boxed{4.28\times10^{-12}\ \text{J}}$$

(b) The Sun is comprised of

$$N = \frac{1.99\times10^{30}\ \text{kg}}{1.67\times10^{-27}\ \text{kg/atom}} = \boxed{1.19\times10^{57}\ \text{atoms}}$$

$$= 1.19\times10^{57}\ \text{protons}$$

(c) The energy that could be created by this many protons in this reaction is

$$(1.19\times10^{57}\ \text{protons})\left(\frac{4.28\times10^{-12}\ \text{J}}{4\ \text{protons}}\right) = 1.27\times10^{45}$$

Then, since $P = \dfrac{E}{\Delta t}$,

$$\Delta t = \frac{E}{P} = \frac{1.27\times10^{45}\ \text{J}}{3.85\times10^{26}\ \text{W}} = 3.31\times10^{18}\ \text{s} = \boxed{105\ \text{billion years}}$$

(d) The time interval in (c) is an order of magnitude larger than the expected remaining lifetime of the Sun. Only the hydrogen in a relatively small core is available as a nuclear fuel. Only in the core are temperatures and densities high enough for the fusion reaction to be self-sustaining.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

ANSWERS TO EVEN-NUMBERED PROBLEMS

P44.2 (a) 68; (b) $^{68}_{30}\mathrm{Zn}$; (c) Isotopes of other elements to the left and right of zinc in the periodic table (from manganese to bromine) may have the same mass number.

P44.4 ~10^{28} protons; (b) 10^{28} neutrons; (c) ~10^{28} electrons

P44.6 184 m

P44.8 (a) 0.360 MeV; (b) Figure P44.8 shows the highest point in the curve at about 4 MeV, a factor of ten higher than the value in (a).

P44.10 $r_2 = \sqrt{\frac{m_2}{m_1}} r_1$

P44.12 (a) 5.18 fm; (b) λ is much less than the distance of closest approach

P44.14 (a) 2.82×10^{-5}; (b) 1.38×10^{-14}

P44.16 0.210 MeV

P44.18 See P44.18 for full explanation.

P44.20 (a) 84.2 MeV; (b) 342 MeV; (c) The nuclear force is so strong that the binding energy greatly exceeds the minimum energy needed to overcome electrostatic repulsion.

P44.22 ~200 MeV

P44.24 (a) Nucleons on the surface have fewer neighbors with which to interact. The surface term is negative to reduce the estimate from the volume term, which assumes that all nucleons have the same number of neighbors; (b) sphere, $\frac{1}{3}r$, cube, $\frac{1}{6}L$. The sphere has a larger ratio to its characteristic length, so it would represent a larger binding energy and be more plausible for a nuclear shape.

P44.26 (a) $1.55 \times 10^{-5}\ \mathrm{s}^{-1}$; (b) 12.4 h; (c) 2.39×10^{13} atoms; (d) 1.88 mCi

P44.28 See P44.28 for full explanation.

P44.30 $\frac{R_0 T_{1/2}}{\ln 2}\left(2^{-t_1/T_{1/2}} - 2^{-t_2/T_{1/2}}\right)$

P44.32 (a) 0.755; (b) 0.570; (c) 9.766×10^{-4}; (d) No. The decay model depends on large numbers of nuclei. After some long but finite time, only one undecayed nucleus will remain. It is likely that the decay of this final nucleus will occur before infinite time.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P44.34 (a) See P44.34(a) for full explanation; (b) See P44.34(b) for full explanation; (c) See ANS. FIG. P44.34(c); (d) 10.9 min; (e) $t_m = \dfrac{\ln\left(\lambda_1/\lambda_2\right)}{\lambda_1 - \lambda_2}$; (f) 10.9 min

P44.36 (a) See P44.36(a) for full explanation; (b) 18.6 keV

P44.38 (a) $^{65}_{28}\text{Ni}^{*}$; (b) $^{211}_{82}\text{Pb}$; (c) $^{55}_{27}\text{Co}$

P44.40 (a) $N_d = N_{P,0} - N_{P,0}\,e^{-\lambda t} = \left(1.00 \times 10^6\right)\left(1 - e^{-0.0693t}\right)$, where t is in hours; (b) The number of daughter nuclei starts from zero at $t = 0$. The number of stable product nuclei always increases with time and asymptotically approaches 1.00×10^6 as t increases without limit; (c) The minimum number of daughter nuclei is zero at $t = 0$. The maximum number of daughter nuclei asymptotically approaches 1.00×10^6 as t increases without limit; (d) The rate of change has its maximum value, $6.93 \times 10^4\ \text{h}^{-1}$, at $t = 0$, after which the rate decreases more and more, approaching zero as t increases without limit.

P44.42 (a) 1.05×10^{21}; (b) 1.37×10^{9}; (c) $3.83 \times 10^{-12}\ \text{s}^{-1}$; (d) 3.17×10^{3} decays/week; (e) 951 decays/week; (f) 9.95×10^{3} yr

P44.44 (a) 0.281; (b) 1.65×10^{-29}; (c) Radon is continuously created.

P44.46 (a) 4.00×10^{9} yr; (b) $\dfrac{N}{N'} = 0.019\,9$ ^{235}U to ^{207}Pb chain and $\dfrac{N}{N'} = 4.60$ for the ^{232}Th to ^{208}Pb chain

P44.48 (a) 5.70 MeV; (b) 3.27 MeV; (c) exothermic

P44.50 (a) $^{197}_{79}\text{Au} + {}^{1}_{0}\text{n} \rightarrow {}^{198}_{79}\text{Au}^{*} \rightarrow {}^{198}_{80}\text{Hg} + {}^{0}_{-1}\text{e} + \bar{\nu}$; (b) 7.89 MeV

P44.52 See ANS. FIG. P44.52(a) and (b).

P44.54 4.42×10^{3} yr

P44.56 While electric charge is conserved (5 + 2 = 1 + 6), the number of nucleons is not (10 + 4 ≠ 1 + 12). Therefore, this reaction cannot occur.

P44.58 (a) The process cannot occur because energy input would be required; (b) Required energy can come from the electrostatic repulsion; (c) 1.20 MeV

P44.60 The Q value of this hypothetical decay is calculated to be –7.62 MeV, which means you would have to add this much energy to the ^{238}U nucleus to make it emit a proton.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P44.62 The nucleus ^{57}Co cannot decay by e^+ emission because the Q value is –0.187 MeV.

P44.64 (a) 61.8 Bq/L; (b) 40.3 d

P44.66 (a) See ANS. FIG. P44.66; (b) $4.17 \times 10^{-3}\ \text{min}^{-1}$, 2.77 h; (c) 4.02×10^{3} counts/min ; (d) 9.65×10^{6} atoms

P44.68 (a) 3.91×10^{9} yr ; (b) no older

P44.70 (a) $\sim 10^{-1\,363}$; (b) 0.891

P44.72 2.64 min

P44.74 (a) See P44.74(a) for full explanation; (b) 1.53 MeV

P44.76 $$\frac{T_Y \ln 2}{\ln\left[2\left(r_1/r_2\right)^{T_Y/\Delta t}\right]}$$

P44.78 (a) 4.28×10^{-12} J; (b) 1.19×10^{57} atoms; (c) 105 billion years; (d) The time interval in (c) is an order of magnitude larger than the expected remaining lifetime of the Sun. Only the hydrogen in a relatively small core is available as a nuclear fuel. Only in the core are temperatures and densities high enough for the fusion reaction to be self-sustaining.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.